PROVEIT RESEARCH CONTINUATION

Exact Harmonic-Tail Identities

Gap-Four Reconstruction, Hurwitz Shifts, and Alternating Moments

Prepared for Vladimir Reshetnikov

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Abstract. We continue the exact harmonic-tail programme in the ProveIt polylogarithm manuscripts. For every pair of even orders with $q - p = 4$, the first q centered moments generate the full specified formal coordinate space. The proof gives an explicit nonzero determinant and a reconstruction requiring only a two-by-two inverse. Its central comparison theorem proves that every generated minor of order two in the fundamental recurrence is strictly negative for arbitrary even gaps. A concrete new $(8, 12)$ identity is also certified by a finite rational telescoper.

We then develop exact shifted Hurwitz-tail moments, including the Bernoulli corrections that distinguish ordinary subtracted sums from continued Dirichlet values. An elementary trigamma-square telescoper gives a free-shift identity and an all-order differentiated family in ordinary zeta values. Spectral coefficients retain the required generalized Stieltjes terms. Finally, signed tail products lead to alternating moment reductions, a two-parameter digamma generating identity, and anchored polylogarithmic primitives. All analytic statements carry explicit convergence or continuation conventions. The accompanying programs check finite rational identities and independent numerical representations; the infinite-family claims are proved in the text.

Repository baseline: `446dca9bd65c20e409d1d2561d4ffad408103e2a`.
The manuscript source files and all six incoming archives were fixed to this revision. The new conclusions are separated from inherited results and from questions that remain open.

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1 The questions advanced in this continuation

The harmonic tails

$$A_p(n) = \zeta(p) - H_n^{(p)} = \zeta(p, n+1), \qquad H_n^{(p)} = \sum_{k=1}^{n} k^{-p}, \quad H_0^{(p)} = 0,$$

connect three languages used throughout the existing programme: ordinary sums involving harmonic numbers, polygamma functions at integer or shifted arguments, and multiple zeta or polylogarithm values. A polynomial weight can destroy ordinary convergence without destroying the exact identity. The useful question is then to specify the finite subtraction and its constant, and to determine which finite collection of moments already contains all the information.

The present article concentrates on this exact-identity question. Asymptotic expansions enter only to justify convergence, constant extraction, and limiting operations. The main result settles one additional infinite family of the existing optimal-prefix conjecture. The other results produce explicit shift, twist, derivative, and primitive formulas around the same tail functions.

1.1 The precise change in the conjecture

For even $2 \le p \le q$ consider the continued centered moments

$$D_{p,q}(-2r), \qquad D_{p,q}(s) = \sum_{n \ge 0} A_p(n) A_q(n) (n + \tfrac{1}{2})^{-s}.$$

The source reports supply q formal coordinates and prove that some finite initial segment generates them. Their stronger conjecture asks whether the first q moments always suffice. At the pinned revision the following families have proofs in the source reports: the diagonal $p = q$, the adjacent case $q - p = 2$, and each fixed smaller order $p = 2, 4, 6$ [2, 3, 4, 5].

Theorem 3.3 proves the entire family $q - p = 4$. Its additional cases begin at $(p, q) = (8, 12)$, since the first three pairs overlap the fixed-order results. The proof is an all-order nonvanishing argument. A table of computed determinants is only a normalization check. Theorem 3.2 supplies the underlying comparison theorem for every even gap, while the text explains why that theorem alone does not settle higher-dimensional determinants.

The rank is taken over $\mathbb{Q}$ on the source's specified formal labels. It yields exact reconstruction identities for the corresponding numbers, whether or not those numbers have further arithmetic relations. It does not assert linear independence of zeta values.

Direction	Earlier status	Contribution here
Optimal moment prefix	Diagonal, gap two, and $p = 2, 4, 6$ proved	Gap four proved at every order; explicit reconstruction and sums
Hurwitz shift	Requested extension of half-integer centering	Finite endpoint and Bernoulli corrections; free-shift trigamma identity
Logarithmic moments	Subtraction constants must be tracked	All spectral coefficients, including Stieltjes resonance terms
Character twists	Proposed exact reduction programme	Signed centered moments, digamma generator, polylogarithmic primitives
Gaussian S_6, revised S_8	Numerical candidates with unresolved proofs	Remain open; no claim of resolution

1.2 Source scope and established mathematics

The repository snapshot is [1]. The current incoming directory contains six reports: *Bernoulli–Tornheim–Digamma*, *Cubic Tail Descent*, *Determinantal Harmonic Resonance*, *Differential Polygamma and Lerch Jets*, *Multishift Reflection*, and *Polygamma Tail Identities*. The source manifest records the Git blob of every acquired manuscript text and archive. We inspected the claims used here and their nearby research agendas; this is a targeted proof audit, not a certification of every theorem in the collective volumes.

Several ingredients are established mathematics. Hoffman's treatment of products of Riemann-zeta tails explains their relation to multiple zeta values and higher products [9]. Teo proves the odd-weight reduction of alternating double Euler sums [10]. The Hurwitz recurrence, its relation to polygamma, the Bernoulli difference law, and Euler–Maclaurin continuation are classical [13]. Bernoulli Hankel determinants also have an extensive literature [12]. These ingredients are credited where used. The additional contribution is the specified determinant proof, the normalized extensions, and their explicit consequences relative to the inspected programme. An exhaustive historical-priority claim is not made.

1.3 Conventions needed for reproducibility

Bernoulli numbers use $B_1 = -1/2$, and Bernoulli polynomials satisfy

$$\frac{te^{xt}}{e^t - 1} = \sum_{m\ge 0} B_m(x)\frac{t^m}{m!}.$$

The multiple-polylogarithm convention uses decreasing indices:

$$\mathrm{Li}_{s_1,\dots,s_d}(z_1,\dots,z_d) = \sum_{n_1>\cdots>n_d\ge 1} \frac{z_1^{n_1}\cdots z_d^{n_d}}{n_1^{s_1}\cdots n_d^{s_d}}.$$

In particular, a one-variable depth-two expression $\mathrm{Li}_{p,q}(z,1)$ weights the larger index. The generalized Stieltjes constants use

$$\zeta(1+s,a) = \frac{1}{s} + \sum_{j\ge 0} \frac{(-1)^j \gamma_j(a)}{j!} s^j, \qquad \gamma_0(a) = -\psi(a).$$

A superscript $\zeta^{(j)}$ denotes a derivative in the spectral variable. Derivatives of ψ denote derivatives in its argument. Every finite part and every convergent subtraction is specified at its point of use. No divergent series is assigned a value without a stated continuation or subtraction rule.

2 Centered moments and their exact normalization

We first establish the analytic meaning of the moment values used in the reconstruction theorem. The finite coordinate formulas are inherited from the tail reports; their normalization is reproduced here so that the additional determinant argument can be checked without reconstructing an earlier archive.

2.1 Centered asymptotic subtraction

Put $x_n = n + 1/2$. For an integer $p \ge 2$, the centered Hurwitz expansion is

$$A_p(n) \sim \sum_{j\ge 0} a_j(p) x_n^{1-p-2j}, \qquad a_0(p) = \frac{1}{p-1}, \quad a_j(p) = \frac{B_{2j}(1/2)(p)_{2j-1}}{(2j)!} \quad (j \ge 1). \tag{2.1}$$

Here $(p)_m = p(p+1)\cdots(p+m-1)$ is the rising factorial. For each fixed truncation the usual Euler–Maclaurin remainder justifies this expansion, locally uniformly in any parameters kept away from their poles [13]. In particular, if p, q are even,

$$A_p(n)A_q(n) \sim \sum_{j\geq 0} c_j(p,q)x_n^{2-p-q-2j}, \qquad c_j(p,q) = \sum_{h=0}^{j} a_h(p)a_{j-h}(q). \tag{2.2}$$

Proposition 2.1 (Continued and ordinary centered moments)**.** *For even $p, q \geq 2$, $D_{p,q}(s)$ initially converges absolutely for $\Re s > 3-p-q$ and has a meromorphic continuation to the plane. Its possible poles are among*

$$s = 3-p-q-2j, \qquad j \geq 0.$$

Every $s = -2r$, $r \geq 0$, is regular. Moreover, with $J = r+1-(p+q)/2$, the exact ordinary identity is

$$D_{p,q}(-2r) = \sum_{n\geq 0}\left\{ x_n^{2r}A_p(n)A_q(n) - \sum_{j=0}^{J} c_j(p,q)x_n^{2r+2-p-q-2j}\right\}. \tag{2.3}$$

The inner sum is empty when $J < 0$. The complete summand is $O(x_n^{-2})$ when subtraction is needed; when it is not needed the original series is absolutely convergent.

Proof. Subtract L terms of (2.2). Summing them back gives

$$\begin{aligned} D_{p,q}(s) = &\sum_{n\geq 0}\left(A_p(n)A_q(n) - \sum_{j=0}^{L-1} c_j x_n^{2-p-q-2j}\right)x_n^{-s} \\ &+ \sum_{j=0}^{L-1} c_j \zeta(s+p+q-2+2j, 1/2). \end{aligned} \tag{2.4}$$

The first term converges normally for $\Re s > 3-p-q-2L$. Increasing L gives the continuation and the listed possible poles. Their positions are odd integers, so none equals $-2r$. At this value choose exactly the nonnegative powers in the polynomial subtraction. Each added-back Hurwitz value is zero because

$$\zeta(-2h, 1/2) = -\frac{B_{2h+1}(1/2)}{2h+1} = 0 \qquad (h \geq 0).$$

This proves (2.3). The statement uses a convergent sum of the complete braces; the separately divergent pieces are never summed on their own. □

The last vanishing is special to the chosen center. It also cannot be differentiated as though it were an identity in the spectral variable: for example $\zeta'(0, 1/2) = -\frac{1}{2}\log 2 \neq 0$. The shifted and logarithmic formulas below preserve the corresponding finite correction terms.

2.2 Finite summation and the block transform

Let CT_N denote the coefficient of $N^0(\log N)^0$ in a specified large-N expansion. Centered even power sums satisfy

$$S_r(N) = \sum_{n=0}^{N-1}(n+1/2)^{2r} = \frac{B_{2r+1}(N+1/2)}{2r+1} = \sum_{j=0}^{r} T_{rj}N^{2j+1}, \tag{2.5}$$

where

$$T_{rj} = \frac{\binom{2r}{2j}}{2j+1}B_{2r-2j}(1/2).$$

The polynomial has zero constant term. By Proposition 2.1, the moment is consequently the constant term of its finite partial sum. The recurrence

$$A_p(k-1) = A_p(k) + k^{-p}$$

and finite summation by parts give

$$D_{p,q}(-2r) = \sum_{j=0}^{r} T_{rj} C_{p,q}(2j+1), \tag{2.6}$$

where, for a positive odd d,

$$C_{p,q}(d) = \mathrm{CT}_N \Big\{ N^d A_p(N) A_q(N) + \sum_{k=1}^{N} k^{d-q} A_p(k) + \sum_{k=1}^{N} k^{d-p} A_q(k) + \sum_{k=1}^{N} k^{d-p-q} \Big\}. \tag{2.7}$$

This identity is obtained at a finite cutoff before any limiting operation. It is also the convenient starting point for shifted and twisted extensions.

For later reference we state the explicit inherited evaluation. Set $W = p + q$, and define

$$U_{p,q}(d) = \begin{cases} \zeta(p, q-d) + \frac{1}{2}\zeta(W-d), & d < q, \\ \displaystyle\sum_{h=1}^{\min\{(d-q+1)/2,\, p/2-1\}} \frac{\binom{d-q+1}{2h}}{d-q+1} B_{d-q+1-2h}\, \zeta(p-2h), & d > q, \end{cases} \tag{2.8}$$

$$R_{p,q}(d) = \begin{cases} 0, & d < W-1, \\ -\dfrac{B_{d-W+1}}{2} \left\{ \dfrac{1}{p}\binom{d-q}{p-1} + \dfrac{1}{q}\binom{d-p}{q-1} \right\}, & d \geq W-1. \end{cases} \tag{2.9}$$

Then

$$C_{p,q}(d) = U_{p,q}(d) + U_{q,p}(d) + R_{p,q}(d). \tag{2.10}$$

All double-zeta sums appearing in (2.8) are convergent. There is no case $d = q$ because d is odd and q is even.

Why these finite formulas hold. For $d < q$, changing the order in the convergent sum yields $\sum_{k\geq 1} k^{d-q} A_p(k) = \zeta(p, q-d)$. For $d > q$ set $m = d - q$ and $F_m(N) = (B_{m+1}(N+1) - B_{m+1})/(m+1)$. The finite identity

$$\sum_{k=1}^{N} k^m A_p(k) = A_p(N) F_m(N) + \sum_{\ell=1}^{N} \frac{F_m(\ell-1)}{\ell^p}$$

reduces everything to Bernoulli polynomials and ordinary power sums. The even Bernoulli terms give (2.8). The half-leading Faulhaber terms leave precisely one half of $\zeta(W-d)$ for each convergent ordered block. When both ordered blocks require reduction, their two half-terms cancel the last sum in (2.7). At $d = W-1$ this includes cancellation of the two $-\gamma/2$ constants against $+\gamma$ from the harmonic sum. The remaining boundary products give (2.9). This is the finite-sum proof in the incoming tail calculus [2, 4]; no additional relation among numerical periods is used.

2.3 Inverse transform and a small check

Write $C_j = C_{p,q}(2j+1)$ and $D_r = D_{p,q}(-2r)$. The Bernoulli generating identity in (2.5) yields the inverse transform

$$C_j = \sum_{r=0}^{j} \binom{2j+1}{2r} 2^{-2(j-r)} D_r. \tag{2.11}$$

One way to verify it is to set $G(t) = \sum D_r t^{2r}/(2r)!$ and $C(t) = \sum C_j t^{2j+1}/(2j+1)!$. The forward transform says $G(t) = C(t)/(2\sinh(t/2))$ as a formal power series. Multiplying back proves (2.11), with no analytic assumption on the growth of the moments.

For $p = q = 2$, the first two blocks are $C_0 = 3\zeta(3)$ and $C_1 = -1/2$. Hence

$$D_0 = 3\zeta(3), \qquad D_1 = -\frac{1}{4}\zeta(3) - \frac{1}{6},$$

and Proposition 2.1 gives the ordinary identity

$$\sum_{n\geq 0} \left((n+1/2)^2 \psi'(n+1)^2 - 1\right) = -\frac{1}{4}\zeta(3) - \frac{1}{6}.$$

These are inherited checks. The next section uses the same fixed normalization to prove an additional infinite reconstruction family.

3 The optimal initial prefix when the even orders differ by four

The current incoming reports prove the optimal initial-prefix statement for the diagonal, for adjacent even orders, and for the fixed smaller orders $p = 2, 4, 6$ [2, 3, 4, 5]. We prove the next fixed-gap family

$$(p, q) = (2b-4, 2b), \qquad b \geq 3.$$

Its additional range relative to those reports begins with $(p, q) = (8, 12)$ and continues with $(10, 14), (12, 16), \ldots$. The proof uses the strict log-convexity of the positive coefficients of $2/x - \cot(x/2)$ to show that a specified two-by-two boundary determinant never vanishes. We also prove a structural statement about every pair of columns in the fundamental recurrence for an arbitrary even gap. The latter does not settle the higher-dimensional boundary determinant.

3.1 Normalization and the determinant to be proved nonzero

For even $2 \leq p \leq q = 2b$, put

$$A_p(n) = \zeta(p) - H_n^{(p)} = \zeta(p, n+1), \qquad D_{p,q}(s) = \sum_{n\geq 0} A_p(n) A_q(n) (n+\tfrac{1}{2})^{-s}.$$

The series initially converges for $\Re s > 3 - p - q$, and its continuation is regular at every $s = -2r$. Write $D_r = D_{p,q}(-2r)$. We use the inherited blocks C_j, characterized by the mutually inverse formulas

$$D_r = \sum_{j=0}^{r} \frac{\binom{2r}{2j}}{2j+1} B_{2r-2j}(\tfrac{1}{2}) C_j, \tag{3.1}$$

$$C_j = \sum_{r=0}^{j} \binom{2j+1}{2r} 2^{-2(j-r)} D_r. \tag{3.2}$$

For $j \geq b$ they have the finite coordinate form

$$C_j = \sum_{k=0}^{b-1} a_{jk} Z_k, \qquad Z_k = \zeta(2k), \qquad Z_0 = -\tfrac{1}{2}.$$

For $d = 2j+1$ and $e = d-p-q+2k+1$, the inherited coefficients are

$$a_{jk} = \begin{cases} B_e \left\{ \dfrac{\binom{d-q+1}{p-2k}}{d-q+1} + \dfrac{\binom{d-p+1}{q-2k}}{d-p+1} \right\}, & e \geq 0, \\ 0, & e < 0. \end{cases} \tag{3.3}$$

Each binomial summand is omitted unless its lower index lies between 2 and its upper index. In particular a zero lower index is omitted. Let

$$M_{p,q} = (a_{b+r-1,k})_{1 \leq r \leq b,\ 0 \leq k < b}. \tag{3.4}$$

The first q moments have full rank on the specified formal labels

$$C_0, \ldots, C_{b-1}, Z_0, \ldots, Z_{b-1}$$

if and only if $M_{p,q}$ is nonsingular. The complete first-q determinant in this coordinate order is $\det M_{p,q}/(4b-1)!!$, by the triangular transform (3.1). No arithmetic independence of the actual constants is asserted by this formal rank statement.

3.2 A strict comparison theorem for all even gaps

Put $q - p = 2c$, where $1 \leq c < b$, and define

$$\alpha_n = \frac{2|B_{2n}|}{(2n)!} = 4 \sum_{m=1}^{\infty} (2\pi m)^{-2n}, \qquad n \geq 1. \tag{3.5}$$

For $s \geq c+1$ set

$$A_s = \frac{(2s)!}{(2s-2c)!}, \qquad B_s = \frac{(2s-1)!}{(2s-2c-1)!}, \qquad d_s = 2(A_s + B_s), \tag{3.6}$$

and put $A_s = 0$ for $1 \leq s \leq c$. The letters A_s, B_s in (3.6) are rational recurrence coefficients, distinguished from the harmonic tails $A_p(n)$ and Bernoulli numbers B_{2n} by their arguments and subscripts. Define

$$K(s,u) = \frac{\alpha_{s-u}(B_s + A_u)}{d_s}, \qquad s \geq c+1, \quad 1 \leq u < s, \tag{3.7}$$

and the fundamental array

$$\begin{aligned} R_{s,j} &= \delta_{sj}, && 1 \leq s, j \leq c, \\ R_{s,j} &= \sum_{u=1}^{s-1} K(s,u) R_{u,j}, && s \geq c+1, \quad 1 \leq j \leq c. \end{aligned} \tag{3.8}$$

All generated entries $R_{s,j}$, $s \geq c+1$, are strictly positive. The earlier fixed-gap reduction gives

$$\det M_{2(b-c),2b} = \frac{(-1)^{b(b-1)/2}}{4^b b!} \left(\prod_{s=c+1}^{b+c} d_s \right) \det(R_{b+i,j})_{1 \leq i,j \leq c}. \tag{3.9}$$

We give the coefficient mechanism below to check the convention and signs directly; nonvanishing of the last determinant is the additional issue addressed here when $c = 2$.

Lemma 3.1 (Strictly decreasing kernel ratios)**.** *For every $s \geq c+1$, the quotient*

$$\frac{K(s+1,u)}{K(s,u)}$$

is a strictly decreasing function of the integer u, $1 \leq u < s$.

Proof. The coefficients in (3.5) are strictly log-convex. Indeed, with $x_m = (2\pi m)^{-2}$,

$$\alpha_n \alpha_{n+2} - \alpha_{n+1}^2 = 16 \sum_{m<\ell} x_m^n x_\ell^n (x_m - x_\ell)^2 > 0.$$

The sums converge absolutely for $n \geq 1$, and the positive support contains more than one point. Thus α_{n+1}/α_n strictly increases with n. On the other hand,

$$\frac{K(s+1,u)}{K(s,u)} = \frac{d_s}{d_{s+1}} \frac{\alpha_{s+1-u}}{\alpha_{s-u}} \frac{B_{s+1} + A_u}{B_s + A_u}.$$

The middle factor strictly decreases with u. The last factor is nonincreasing because A_u is nondecreasing, $B_{s+1} > B_s > 0$, and $(B_{s+1} + a)/(B_s + a)$ decreases as a increases. Every factor is positive, so their product strictly decreases. □

Theorem 3.2 (Every generated minor of order two is negative)**.** *Fix an even gap $2c$ with $c \geq 2$. For any initial columns $1 \leq j < k \leq c$, the positive ratios*

$$\frac{R_{s,k}}{R_{s,j}}, \qquad s \geq c+1,$$

strictly decrease with s. Equivalently, for $c+1 \leq r < s$,

$$\det \begin{pmatrix} R_{r,j} & R_{r,k} \\ R_{s,j} & R_{s,k} \end{pmatrix} < 0. \tag{3.10}$$

Proof. Write $X_u = R_{u,j}$, $Y_u = R_{u,k}$. Suppose that the stated ratio ordering has been proved through row $s \geq c+1$, and put $z_s = Y_s/X_s$. This supposition is empty at $s = c+1$, so it includes the induction base. For $u < s$ set $v_u = Y_u - z_s X_u$. Among the initial rows, $v_j = -z_s < 0$, $v_k = 1 > 0$, and all other v_u vanish. Among the generated rows $c+1 \leq u < s$, the induction hypothesis gives $v_u > 0$. Thus the unique negative term is at index j, and every positive term has larger index. The recurrence gives

$$\sum_{u=1}^{s-1} K(s,u) v_u = 0.$$

Put $\beta_u = K(s+1,u)/K(s,u)$. Since $v_s = 0$,

$$\begin{aligned} Y_{s+1} - z_s X_{s+1} &= \sum_{u=1}^{s-1} K(s,u) \beta_u v_u \\ &= \sum_{u=1}^{s-1} K(s,u)(\beta_u - \beta_j) v_u < 0. \end{aligned}$$

The negative-index term vanishes; every nonzero remaining term is negative by Lemma 3.1. Strictness already follows from the term $u = k$. Hence $Y_{s+1}/X_{s+1} < z_s$, completing the induction. Multiplying the ratio inequality by the two positive denominators proves (3.10). □

The one exceptional initial negative index is essential in this proof. Positivity of the recurrence entries alone would not imply the theorem.

3.3 Exact determinant and reconstruction for gap four

Now take $c = 2$. Define $X_s = R_{s,1}$ and $Y_s = R_{s,2}$, so

$$(X_1, Y_1) = (1, 0), \qquad (X_2, Y_2) = (0, 1),$$

and compute both columns by (3.8). In this case

$$d_s = 16(s-1)^2(2s-1)(2s-3), \qquad s \geq 3. \tag{3.11}$$

Set

$$E_s = X_{s+1}Y_s - X_sY_{s+1}, \qquad s \geq 3. \tag{3.12}$$

Theorem 3.2 proves $E_s > 0$ at every index. For example,

$$E_3 = \frac{1}{14515200}, \qquad E_4 = \frac{61}{1170505728000}, \qquad E_5 = \frac{3419}{69528040243200000}.$$

These are exact rational recurrence values, not approximations.

Theorem 3.3 (Optimal prefix for every gap-four pair)**.** *For every integer $b \geq 3$,*

$$\boxed{\det M_{2b-4,2b} = (-1)^{b(b-1)/2+1} \frac{4^b((b+1)!)^2(2b+3)!!(2b+1)!!}{3b!} E_{b+1} \neq 0.} \tag{3.13}$$

Consequently, for $(p, q) = (2b-4, 2b)$,

$$\begin{aligned} \operatorname{span}_{\mathbb{Q}}\{D_0, D_1, \ldots\} &= \operatorname{span}_{\mathbb{Q}}\{D_0, \ldots, D_{q-1}\} \\ &= \operatorname{span}_{\mathbb{Q}}\{C_0, \ldots, C_{b-1}, Z_0, \ldots, Z_{b-1}\}. \end{aligned} \tag{3.14}$$

The complete first-q coefficient determinant is (3.13) divided by $(4b-1)!!$. The prefix is optimal for rank q on these q formal labels.

Proof. The last determinant in (3.9) is now

$$\det \begin{pmatrix} X_{b+1} & Y_{b+1} \\ X_{b+2} & Y_{b+2} \end{pmatrix} = -E_{b+1} < 0.$$

The elementary product evaluation

$$\prod_{s=3}^{b+2} d_s = \frac{16^b((b+1)!)^2(2b+3)!!(2b+1)!!}{3}$$

gives (3.13). The block decomposition and invertible triangular transforms give (3.14). Fewer than q rows cannot have rank q on q specified formal coordinates. □

The first five determinants are

(p, q)	$\det M_{p,q}$
$(2, 6)$	$61/5760$
$(4, 8)$	$-3419/138240$
$(6, 10)$	$-713039/6451200$
$(8, 12)$	$263589503/309657600$
$(10, 14)$	$818219064067/78033715200$

The first three pairs overlap the previously settled fixed smaller orders. The last two belong to the additional infinite range covered by Theorem 3.3.

Checking the source reduction and the rotation. For an arbitrary coefficient vector $v = (v_0, \dots, v_{b-1})$, set $Q(t) = \sum_{k=0}^{b-1} v_k t^{2b-2k}/(2b-2k)!$ and write $Q(ix) = R(x) = \sum_{u=1}^{b} r_u x^{2u}$. Thus $r_u = (-1)^u v_{b-u}/(2u)!$. Consider the real odd expression

$$(R^{(2c)} - R^{(2c)}(0)) \cot(x/2) + (R \cot(x/2))^{(2c)}.$$

The coefficient at x^{2r-1}, with $s = r + c$, is

$$Y_r^* = d_s r_s - \sum_{u=1}^{s-1} \alpha_{s-u}(B_s + A_u) r_u, \qquad r_{b+1} = \cdots = r_{b+c} = 0. \tag{3.15}$$

Indeed $2/x$ supplies d_s, and the remaining cotangent coefficients are $-\alpha_n$. The constant derivative subtraction sets $A_c = 0$, exactly enforcing the omitted zero lower binomial index. Comparing factorials directly with (3.3) gives the row scaling

$$Y_r^* = \frac{2(-1)^{r-c}}{(2r-1)!} (M_{p,q} v)_r.$$

Here is a complete determinant check of (3.9). Let L be the lower triangular matrix of size $b + c$ whose first c rows are the corresponding identity rows, and whose remaining entries are

$$L_{ss} = d_s, \qquad L_{su} = -d_s K(s,u) \ (u < s), \qquad L_{su} = 0 \ (u > s).$$

Its inverse has entries $(L^{-1})_{s,j} = R_{s,j}$ in its first c columns. The coefficient matrix H of (3.15) is the minor with rows $c+1, \dots, b+c$ and columns $1, \dots, b$. Jacobi's complementary minor identity therefore gives

$$\det H = (-1)^{bc} \left(\prod_{s=c+1}^{b+c} d_s \right) \det(R_{b+i,j})_{1 \le i,j \le c}.$$

If T denotes the reversed diagonal map from v to r and S the row scaling just displayed, then $HT = SM_{p,q}$ and

$$\det T = \frac{(-1)^b}{\prod_{u=1}^{b} (2u)!}, \qquad \det S = \frac{2^b (-1)^{b(b+1)/2 - bc}}{\prod_{r=1}^{b} (2r-1)!}.$$

Taking determinants yields exactly (3.9). This checks the recurrence against the original Bernoulli matrix, including the sign convention, without a new analytic assumption.

For $c = 2$, these equations give a particularly small reconstruction. Given the actual blocks $C_b, \dots, C_{2b-1}$, set

$$Y_r^* = \frac{2(-1)^{r-2}}{(2r-1)!} C_{b+r-1}, \qquad 1 \le r \le b.$$

Set $\eta_1 = \eta_2 = 0$ and compute, for $3 \le s \le b+2$,

$$\eta_s = \frac{Y_{s-2}^*}{d_s} + \sum_{u=1}^{s-1} K(s,u) \eta_u. \tag{3.16}$$

The two initial coefficients are recovered by

$$\begin{pmatrix} r_1 \\ r_2 \end{pmatrix} = - \begin{pmatrix} X_{b+1} & Y_{b+1} \\ X_{b+2} & Y_{b+2} \end{pmatrix}^{-1} \begin{pmatrix} \eta_{b+1} \\ \eta_{b+2} \end{pmatrix}. \tag{3.17}$$

Then $r_s = X_s r_1 + Y_s r_2 + \eta_s$ and $Z_k = (-1)^{b-k}(2b-2k)! r_{b-k}$. This uses one two-by-two inverse with a proved nonzero determinant. Equations (3.2) and (3.17) reconstruct all the even-zeta coordinates from the first q moments; substitution into (3.3) and (3.1) recovers every later moment.

3.4 An infinite family of ordinary rational polygamma sums

Write the first row of $M_{p,q}^{-1}$ as $(\lambda_b, \dots, \lambda_{q-1})$ and define

$$w_r = \sum_{j=\max\{b,r\}}^{q-1} \lambda_j \binom{2j+1}{2r} 2^{-2(j-r)}, \qquad P_b(x) = \sum_{r=0}^{q-1} w_r x^{2r}. \tag{3.18}$$

For $p = q-4$, put

$$\begin{aligned} u_0 &= \frac{1}{(p-1)(q-1)}, \\ u_1 &= -\frac{1}{24}\left(\frac{q}{p-1} + \frac{p}{q-1}\right), \\ u_2 &= \frac{7}{5760}\left(\frac{q(q+1)(q+2)}{p-1} + \frac{p(p+1)(p+2)}{q-1}\right) + \frac{pq}{576}, \\ S_b(x) &= u_0 w_{q-1} x^4 + (u_0 w_{q-2} + u_1 w_{q-1}) x^2 + u_0 w_{q-3} + u_1 w_{q-2} + u_2 w_{q-1}. \end{aligned}$$

Corollary 3.4 (Gap-four rational evaluations)**.** *For every even $q \geq 6$, with $p = q-4$ and $x_n = n + 1/2$,*

$$\boxed{\sum_{n=0}^{\infty} \left\{ \frac{P_b(x_n)}{(p-1)!(q-1)!} \psi^{(p-1)}(n+1)\psi^{(q-1)}(n+1) - S_b(x_n) \right\} = -\frac{1}{2}.} \tag{3.19}$$

The sum converges absolutely. Its polynomial P_b has exact degree $2q-2$, because

$$w_{q-1} = \frac{4b(4b-1)E_b}{d_{b+2}E_{b+1}} > 0. \tag{3.20}$$

Proof. The inverse row gives $\sum_{j=b}^{q-1} \lambda_j C_j = Z_0 = -1/2$. The inverse Bernoulli transform gives the weights (3.18). The centered expansion of the tail product is

$$A_p(n)A_q(n) = u_0 x_n^{2-p-q} + u_1 x_n^{-p-q} + u_2 x_n^{-p-q-2} + O(x_n^{-p-q-4}).$$

Since $\deg P_b \leq 2q-2$ and $p = q-4$, its nonnegative powers are exactly $S_b(x_n)$; the remaining summand is $O(x_n^{-2})$. The continued centered sums of $x_n^4, x_n^2, 1$ are all zero, so the ordinary sum equals the moment combination. Equivalently, finite centered power summation gives these polynomial subtraction terms zero constant coefficient at the integer cutoff. The polygamma normalization follows from even p, q.

For (3.20), set every row output in (3.15) except the last equal to zero. A homogeneous solution before the final boundary equation has initial vector proportional to $(Y_{b+1}, -X_{b+1})$, since its coefficient at $b+1$ must vanish. Its coefficient at b is $-E_b$ times that scalar, and its continued coefficient at $b+2$ is E_{b+1} times that scalar. Hence the last row gives

$$\frac{r_b}{Y_b^*} = \frac{E_b}{d_{b+2}E_{b+1}}.$$

The row and column scalings above show that $\lambda_{q-1} = 4bE_b/(d_{b+2}E_{b+1})$. Multiplication by $\binom{2q-1}{2q-2} = 4b-1$ gives the stated positive coefficient. □

3.5 A concrete evaluation at the new pair $(8, 12)$

A centered difference keeps the integer coefficients readable. Put

$$\begin{aligned} K(y) &= y^{13}(469839 + 1269227y^2 - 11819495y^4 \\ &\qquad + 56443363y^6 + 83672540y^8 + 12834702y^{10}), \\ P(x) &= K(x + \tfrac{1}{2}) - K(x - \tfrac{1}{2}), \\ S(x) &= \frac{2951981460x^4 + 51273021735x^2 + 118231913818}{770}. \end{aligned}$$

Then the exact ordinary identity is

$$\boxed{\sum_{n=0}^{\infty} \left\{ \frac{P(n+\frac{1}{2})}{7!\,11!} \psi^{(7)}(n+1)\psi^{(11)}(n+1) - S(n+\tfrac{1}{2}) \right\} = -\frac{790768509}{16}.} \tag{3.21}$$

For example, the coefficient row

$$(469839, 1269227, -11819495, 56443363, 83672540, 12834702)$$

multiplied by $M_{8,12}$ is exactly $(790768509/8, 0, 0, 0, 0, 0)$. The centered difference converts this row of blocks into P, and the first three centered tail-product coefficients give precisely S. Thus (3.21) follows already from the preceding corollary.

There is also a finite telescoping proof, independent of the continued-moment interpretation. Let $c_0, \dots, c_5$ be the six coefficients in the parentheses defining K, in increasing order, and let

$$F_m(N) = \sum_{k=1}^{N} k^m = \frac{B_{m+1}(N+1) - B_{m+1}}{m+1}.$$

Define the explicitly specified polynomials

$$U_8(N) = \sum_{j=0}^{5} c_j F_{2j+1}(N), \qquad U_{12}(N) = \sum_{j=0}^{5} c_j F_{2j+5}(N),$$

$$\begin{aligned} V(N) = -\frac{N}{6160}(&2417202210N^4 - 5764920315N^3 + 42395905270N^2 \\ &- 129829345800N + 260578219219). \end{aligned}$$

The first two definitions imply

$$\Delta U_8(N) = K(N)N^{-12}, \qquad \Delta U_{12}(N) = K(N)N^{-8},$$

where $\Delta f(N) = f(N) - f(N-1)$. Direct polynomial expansion also gives

$$\Delta V(N) = \frac{K(N)}{N^{20}} + \frac{U_8(N-1)}{N^8} + \frac{U_{12}(N-1)}{N^{12}} - S(N - \tfrac{1}{2}). \tag{3.22}$$

In particular the negative powers cancel on the right. These are finite rational-function identities, reproducible from the six integer coefficients and the displayed Bernoulli polynomials. Now put

$$\mathcal{F}(N) = K(N)A_8(N)A_{12}(N) + U_8(N)A_8(N) + U_{12}(N)A_{12}(N) + V(N).$$

Using $A_m(N-1) = A_m(N) + N^{-m}$ and the three difference identities, all mixed terms cancel to give

$$\mathcal{F}(N) - \mathcal{F}(N-1) = P(N - \tfrac{1}{2})A_8(N-1)A_{12}(N-1) - S(N - \tfrac{1}{2}).$$

Since $\mathcal{F}(0) = 0$, the partial sum of (3.21) through $n = N - 1$ is exactly $\mathcal{F}(N)$. For the limiting step, use

$$A_8(N) = \frac{1}{7N^7} - \frac{1}{2N^8} + \frac{2}{3N^9} - \frac{1}{N^{11}} + O(N^{-13}),$$
$$A_{12}(N) = \frac{1}{11N^{11}} - \frac{1}{2N^{12}} + \frac{1}{N^{13}} - \frac{91}{30N^{15}} + O(N^{-17}).$$

The three tail-containing terms of $\mathcal{F}$ have constant coefficients, respectively,

$$-\frac{116518888243}{2310}, \qquad -\frac{193499245}{336}, \qquad \frac{350665911}{220}.$$

Their sum is $-790768509/16$, V has zero constant coefficient, and the coefficients of $N, N^2, \ldots, N^5$ cancel exactly. The remainders above contribute $O(N^{-1})$. This proves the ordinary evaluation by a finite identity and an asymptotic limit.

3.6 Scope and reproducible checks

The exact script checks (3.13) against the original Bernoulli matrix for $3 \le b \le 18$, checks the independent rotation matrix, and verifies (3.20) where a full inverse is computed. It also checks 1,946 minors of order two for gaps $c = 2, \ldots, 6$, and verifies the displayed ordinary evaluation by an exact finite telescoper. These checks detect normalization errors; the all-order proof is Theorem 3.2, not the finite arithmetic range.

The theorem about minors of order two cannot be promoted to a fixed-sign assertion for larger boundary determinants. For example, for $c = 3$ the exact signs are

$$\det(R_{6+i,j})_{1\le i,j\le 3} < 0, \qquad \det(R_{7+i,j})_{1\le i,j\le 3} > 0.$$

Both are nonzero, and both arrays satisfy all the strict negative minor statements in Theorem 3.2. Thus order-two comparison alone does not prove the general conjecture. Combining all currently established families, the still-unsettled range is

$$8 \le p < q, \qquad q - p \ge 6, \qquad p, q \text{ even}.$$

A useful next problem is to identify an exact, sign-flexible nonvanishing invariant for the higher boundary determinants, or to combine the fixed-gap recurrence with the complementary determinant of size $p/2$ from the shifted-Hankel approach. The rank problem and the arithmetic independence of the actual zeta coordinates remain distinct questions.

4 Hurwitz-shifted tails: Bernoulli primitives and exact corrections

The centered-tail results in the inspected reports use the lattice $n + 1/2$ and the tails $\zeta(p, n + 1)$. Their ordinary subtraction rule uses the special vanishing $\zeta(-2j, 1/2) = 0$. A shift changes both the lower endpoint of the finite summation and the continued sums of the subtracted polynomials. The following formulas retain both changes. They answer the exact-reduction part of the proposed Hurwitz-shifted extension; they do not assert a new determinant nonvanishing theorem.

The ingredients are the classical Hurwitz recurrence, the Bernoulli difference law, and Euler–Maclaurin continuation [13, 14]. The particular discrete primitive, shifted block formulas, and normalized identities are proved below without a numerical relation search.

4.1 An explicit discrete primitive at every sufficiently large degree

For an integer s, use the pole-free convention

$$\mathcal{Z}_s(v) = \begin{cases} \zeta(s,v), & s \neq 1, \\ -\psi(v), & s = 1. \end{cases}$$

Thus $\mathcal{Z}_s(v) - \mathcal{Z}_s(v+1) = v^{-s}$ at every integer s, including $s = 1$. For integers $s \geq 2$ and $m \geq 0$, define the finite expression

$$\mathcal{L}_{s,m}(v) = \frac{1}{m+1} \sum_{\substack{0 \leq h \leq m+1 \\ h \text{ even}}} \binom{m+1}{h} B_h \, \mathcal{Z}_{s-m-1+h}(v) - \frac{B_{m+1}(v)}{m+1} \zeta(s,v). \tag{4.1}$$

All nonpositive integer Hurwitz values in this formula are Bernoulli polynomials. In particular the formula is finite even when m is large.

Theorem 4.1 (Bernoulli discrete primitive)**.** *Let $p, q \geq 2$ be integers, let $d \geq \max(p,q)$ be an integer, and let $v > 0$. Put*

$$\mathcal{C}_d^{p,q}(v) = \mathcal{L}_{p,d-q}(v) + \mathcal{L}_{q,d-p}(v), \tag{4.2}$$

$$\mathcal{K}_d^{p,q}(v) = \mathcal{C}_d^{p,q}(v) - (v-1)^d \zeta(p,v)\zeta(q,v). \tag{4.3}$$

Then

$$\boxed{\mathcal{K}_d^{p,q}(v) - \mathcal{K}_d^{p,q}(v+1) = \{v^d - (v-1)^d\}\zeta(p,v)\zeta(q,v).} \tag{4.4}$$

Consequently, for every integer $N \geq 1$,

$$\sum_{n=0}^{N-1} \{(n+v)^d - (n+v-1)^d\}\zeta(p,n+v)\zeta(q,n+v) = \mathcal{K}_d^{p,q}(v) - \mathcal{K}_d^{p,q}(v+N). \tag{4.5}$$

If also $d < p+q-2$, the infinite series is ordinary and absolutely convergent, and its value is $\mathcal{K}_d^{p,q}(v)$.

Proof. Write $R_m(v) = B_{m+1}(v)/(m+1)$. The Bernoulli difference law is $R_m(v+1) - R_m(v) = v^m$. Because $B_1 = -1/2$ and all higher odd Bernoulli numbers vanish,

$$\frac{1}{m+1} \sum_{h \text{ even}} \binom{m+1}{h} B_h v^{m+1-h} = R_m(v) + \frac{1}{2} v^m.$$

Use these two identities together with the Hurwitz recurrence in (4.1). The result is exactly

$$\mathcal{L}_{s,m}(v) - \mathcal{L}_{s,m}(v+1) = v^m \zeta(s,v) - \frac{1}{2} v^{m-s}. \tag{4.6}$$

Thus $\mathcal{C}_d(v) - \mathcal{C}_d(v+1)$ equals

$$v^{d-q}\zeta(p,v) + v^{d-p}\zeta(q,v) - v^{d-p-q} = v^d\{\zeta(p,v)\zeta(q,v) - \zeta(p,v+1)\zeta(q,v+1)\}.$$

Subtracting the difference of $(v-1)^d\zeta(p,v)\zeta(q,v)$ gives (4.4); summing gives (4.5). In the stated convergent range, the summand is $O(n^{d+1-p-q})$. Direct use of the Hurwitz asymptotic expansion in (4.3) gives $\mathcal{K}_d^{p,q}(v) = O(v^{d+2-p-q})$, hence a zero limit at infinity. Alternatively this last bound follows term by term in (4.1), since all its Hurwitz orders are then at least two. □

The convention at $s = 1$ in (4.1) is essential. Inserting $\zeta(1,v)$ into the displayed finite expression would make a valid discrete primitive appear divergent. Its correct coordinate is $\mathcal{Z}_1(v) = \gamma_0(v) = -\psi(v)$.

4.2 Every even centered moment, with its lower endpoint

Assume now that $2 \le p \le q$ are even and $v > 1/2$. Put

$$\alpha = v - \frac{1}{2}, \qquad a = v - 1, \qquad x_n = n + \alpha, \qquad A_p(n; v) = \zeta(p, n + v),$$

and let

$$D_{p,q}(s; v) = \sum_{n \ge 0} A_p(n; v) A_q(n; v) x_n^{-s}, \qquad \Re s > 3 - p - q. \tag{4.7}$$

Its meromorphic continuation is understood outside this half-plane. For $r \ge 0$ set $D_r(v) = D_{p,q}(-2r; v)$ and

$$\widehat{D}_r(v) = D_r(v) - \zeta(-2r, \alpha)\zeta(p, v)\zeta(q, v). \tag{4.8}$$

For $u \ge 2$, $w \ge 1$, use the convergent strict double Hurwitz value

$$Z_v(u, w) = \sum_{\ell > k \ge 0} (\ell + v)^{-u} (k + v)^{-w}.$$

Theorem 4.2 (Complete shifted Bernoulli transform)**.** *Every $D_r(v)$ is regular. Define $\mathcal{C}_d(v)$ for positive odd d by the following three formulas:*

$$\mathcal{C}_d(v) = Z_v(p, q - d) + Z_v(q, p - d) + \zeta(p + q - d, v), \qquad d < p, \tag{4.9}$$

$$\mathcal{C}_d(v) = Z_v(p, q - d) + \frac{1}{2}\zeta(p + q - d, v) + \mathcal{L}_{q,d-p}(v), \qquad p < d < q, \tag{4.10}$$

$$\mathcal{C}_d(v) = \mathcal{L}_{p,d-q}(v) + \mathcal{L}_{q,d-p}(v), \qquad d > q. \tag{4.11}$$

The middle range is empty when $p = q$. Since p, q are even and d is odd, these three ranges cover all required indices. Then

$$\widehat{D}_r(v) = \sum_{j=0}^{r} \frac{\binom{2r}{2j}}{2j + 1} B_{2r-2j}(1/2)\, \mathcal{C}_{2j+1}(v), \tag{4.12}$$

$$\mathcal{C}_{2j+1}(v) = \sum_{r=0}^{j} \binom{2j + 1}{2r} 2^{-2(j-r)} \widehat{D}_r(v). \tag{4.13}$$

At $v = 1$ the correction in (4.8) vanishes. For a general shift it is a specified Bernoulli polynomial multiplied by the initial product of tails.

Proof. The centered expansion is

$$\zeta(p, x + 1/2) \sim x^{1-p} \left\{ \frac{1}{p - 1} + \sum_{h \ge 1} \frac{B_{2h}(1/2)(p)_{2h-1}}{(2h)!} x^{-2h} \right\}. \tag{4.14}$$

Consequently the only possible poles of (4.7) are at $s = 3 - p - q - 2h$, all of them odd integers. Finite subtraction of (4.14) proves regularity at every $-2r$.

Let $Q_r(y) = B_{2r+1}(y + 1/2)/(2r + 1)$. It is odd, and

$$\sum_{n=0}^{N-1} x_n^{2r} = Q_r(N + a) - Q_r(a), \qquad Q_r(a) = -\zeta(-2r, \alpha).$$

Finite summation by parts with $F(t) = \zeta(p, t)\zeta(q, t)$ therefore expresses the partial moment as

$$\sum_{j=0}^{r} \frac{\binom{2r}{2j}}{2j + 1} B_{2r-2j}(1/2) E_{2j+1}(N; v) - Q_r(a) F(v), \tag{4.15}$$

where, with $y = N + v - 1$,

$$E_d(N; v) = y^d F(y+1) + \sum_{k=0}^{N-1} (k+v)^{d-p} \zeta(q, k+v+1) + \sum_{k=0}^{N-1} (k+v)^{d-q} \zeta(p, k+v+1) + \sum_{k=0}^{N-1} (k+v)^{d-p-q}.$$

All equalities so far are finite identities.

Take constant terms as $y \to \infty$. The continued value of the partial moment equals its constant term in this coordinate: every nonnegative asymptotic monomial is an even power of x, and $\zeta(-2h, N+\alpha) = \zeta(-2h, y+1/2)$ is an odd polynomial in y, hence contributes zero constant term at the upper endpoint.

For $d < p$ every sum in E_d converges and the boundary tends to zero, giving (4.9). For $p < d < q$, apply the finite Faulhaber identity

$$\sum_{k=0}^{N-1} (k+v)^m \zeta(s, k+v+1) = \frac{B_{m+1}(y+1) - B_{m+1}(v)}{m+1} \zeta(s, y+1) + \frac{1}{m+1} \sum_{k=0}^{N-1} \frac{B_{m+1}(k+v) - B_{m+1}(v)}{(k+v)^s}$$

to its one nonnegative power. Here $m = d - p$ is odd and $m < q - 1$, so no endpoint term remains. Its $B_1 = -1/2$ term combines with the final sum in E_d to give $\frac{1}{2}\zeta(p+q-d, v)$; the other terms are precisely (4.10).

For $d > q$, Theorem 4.1 gives the exact identity $E_d(N; v) = \mathcal{C}_d(v) - \mathcal{K}_d(y+1)$. For completeness the last endpoint has zero constant term. Write $c_{s,j} = [y^{-j}]\zeta(s, y+1)$. For even s, every exponent in this expansion is odd except the term $-\frac{1}{2}y^{-s}$. Since m is odd, direct constant-term extraction in (4.1) gives

$$\mathrm{CT}_y \, \mathcal{L}_{s,m}(y+1) = -\frac{1}{2} c_{s,m}.$$

Indeed the constant from the first finite sum cancels the even-power part of $B_{m+1}(y+1)\zeta(s, y+1)/(m+1)$; its remaining contribution is $c_{s,m}/2$, from $B_1(1) = 1/2$. On the other hand

$$\mathrm{CT}_y\{y^d F(y+1)\} = -\frac{1}{2}\{c_{p,d-q} + c_{q,d-p}\}.$$

Subtracting proves $\mathrm{CT}_y \, \mathcal{K}_d(y+1) = 0$. This also covers $d = p + q - 1$, where individual harmonic logarithms in finite Faulhaber sums cancel only after all terms are combined. Thus (4.11) holds at every required high index.

Taking constant terms in (4.15) proves (4.12). Finally the Bernoulli polynomial identity

$$y^{2j+1} = \sum_{r=0}^{j} \binom{2j+1}{2r} 2^{-2(j-r)} Q_r(y)$$

inverts the finite transform and gives (4.13). □

For $d > q$, every order in (4.11) is even and no larger than q. Orders below zero are Bernoulli polynomials. It follows that, over $\mathbb{Q}(v)$, all even moments lie in the span of

$$\mathcal{C}_1(v), \mathcal{C}_3(v), \ldots, \mathcal{C}_{q-1}(v), \quad 1, \zeta(2, v), \zeta(4, v), \ldots, \zeta(q, v), \quad \zeta(p, v)\zeta(q, v).$$

This is a list of $q + 2$ specified coordinates, not an assertion of independence. At $v = 1$ the initial-product correction vanishes and the highest single-zeta column cancels, recovering the smaller unshifted coordinate system. At rational v, all displayed coefficients and all negative even Hurwitz values are rational.

4.3 Ordinary sums and every Stieltjes correction

The following statement also covers odd tail orders and odd polynomial weights. Its derivatives are in the Dirichlet weight x_n^{-s}, not in the spectral orders p, q of the two tails.

Theorem 4.3 (Exact subtraction and all weight jets)**.** *Let $p, q \geq 2$ be integers, $v > 1/2$, P a polynomial, and*

$$f_n = P(x_n)\zeta(p, n+v)\zeta(q, n+v).$$

Let $R(X) = \sum_{k\geq 0} r_k X^k$ be the nonnegative-power part of its asymptotic expansion in x_n, and let ρ be the coefficient of x_n^{-1}. These are finite coefficient calculations from (4.14). Set $u_n = f_n - R(x_n) - \rho/x_n$. Then $u_n = O(x_n^{-2})$ and

$$\sum_{n\geq 0} f_n x_n^{-s} = \sum_{n\geq 0} u_n x_n^{-s} + \sum_{k\geq 0} r_k \zeta(s-k, \alpha) + \rho\zeta(1+s, \alpha) \tag{4.16}$$

is a meromorphic identity near $s = 0$, with residue ρ. For every $j \geq 0$,

$$[s^j]\left(\sum_{n\geq 0} f_n x_n^{-s} - \frac{\rho}{s}\right) = \frac{(-1)^j}{j!}\sum_{n\geq 0} u_n \log^j x_n + \frac{1}{j!}\sum_{k\geq 0} r_k \zeta^{(j)}(-k, \alpha) + \frac{(-1)^j \rho}{j!}\gamma_j(\alpha). \tag{4.17}$$

All sums on the right are ordinary and absolutely convergent. In particular, when p, q and P are even, $\rho = 0$ and

$$\boxed{\sum_{n\geq 0}\{f_n - R(x_n)\} = \sum_r [X^{2r}]P(X)\, D_r(v) - \sum_{k\geq 0} r_k \zeta(-k, \alpha).} \tag{4.18}$$

Proof. Subtract the finitely many indicated terms in an initial right half-plane of absolute convergence. The remainder series is normally convergent for $\Re s > -1$, as are all its fixed s derivatives on compact subsets. This proves (4.16) by unique meromorphic continuation. Expand the ordinary Hurwitz terms and

$$\zeta(1+s, \alpha) = \frac{1}{s} + \sum_{j\geq 0} \frac{(-1)^j}{j!}\gamma_j(\alpha) s^j.$$

Coefficient comparison gives (4.17). The even case has only even powers of x_n, so no x_n^{-1} term occurs. Its constant coefficient is (4.18). □

The polynomial correction in (4.18) vanishes on the original lattice $\alpha = 1/2$. It must generally be retained after a shift. For example at $\alpha = 1$, a subtracted constant c contributes $-c\zeta(0, 1) = c/2$, even though all higher negative even zeta values vanish there.

4.4 Explicit trigamma identities and derivative families

Let $T(v) = \psi'(v) = \zeta(2, v)$. Formula (4.2) with $p = q = 2$, $d = 3$ gives

$$\mathcal{C}_3(v) = \frac{1}{2} - v + v(1-v)T(v), \qquad \mathcal{K}_3(v) = \frac{1}{2} - v + v(1-v)T(v) - (v-1)^3 T(v)^2.$$

The summand has constant asymptotic part 3. Applying (4.18) yields the ordinary identity

$$\boxed{\sum_{n\geq 0}\left[\left\{3(n+v-\tfrac{1}{2})^2 + \tfrac{1}{4}\right\}\psi'(n+v)^2 - 3\right] = 2v - \frac{5}{2} + v(1-v)\psi'(v) - (v-1)^3\psi'(v)^2.} \tag{4.19}$$

Initially $v > 1/2$; normal convergence extends this identity to every real $v > 0$, and indeed to $\Re v > 0$. Two exact specializations are

$$\sum_{n\ge 0}\Big[(3n^2+3n+1)(\zeta(2)-H_n^{(2)})^2-3\Big]=-\frac{1}{2}, \tag{4.20}$$

$$\sum_{n\ge 0}\Big[\{3(n+1)^2+\tfrac{1}{4}\}\zeta(2,n+\tfrac{3}{2})^2-3\Big]=\frac{3}{2}+\frac{\pi^2}{8}-\frac{\pi^4}{32}. \tag{4.21}$$

The second follows from $\zeta(2,3/2)=\pi^2/2-4$.

All parameter derivatives are again ordinary convergent identities. For example, with $x_n = n+v-1/2$,

$$\begin{aligned}\sum_{n\ge 0}&\Big[6x_n\psi'(n+v)^2+2(3x_n^2+\tfrac{1}{4})\psi'(n+v)\psi''(n+v)\Big]\\ &=2+(1-2v)\psi'(v)+v(1-v)\psi''(v)-3(v-1)^2\psi'(v)^2-2(v-1)^3\psi'(v)\psi''(v).\end{aligned} \tag{4.22}$$

At $v=1$ this evaluates the displayed mixed polygamma sum to $2-\zeta(2)$. More generally, the mth derivative of the summand in (4.19), for $m\ge 1$, is the explicit finite Leibniz sum

$$\sum_{j=0}^{\min(m,2)}\binom{m}{j}P^{(j)}(x_n)\sum_{k=0}^{m-j}\binom{m-j}{k}\psi^{(k+1)}(n+v)\psi^{(m-j-k+1)}(n+v),\qquad P(X)=3X^2+\tfrac{1}{4}.$$

Its ordinary sum is the corresponding derivative of the right side of (4.19). Normal convergence follows either by differentiated Euler–Maclaurin remainders or by Cauchy estimates on compact subsets of $\Re v > 0$.

Corollary 4.4 (Every derivative at the integral lattice)**.** *Put $P_0(z)=3z^2-3z+1$ and, for $k\ge 1$, define the ordinary sum*

$$S_k=\frac{1}{k!}\sum_{n\ge 0}\frac{d^k}{dz^k}\{P_0(z)\psi'(z)^2\}\Big|_{z=n+1}.$$

Then $S_1=2-\zeta(2)$, and for every $k\ge 2$,

$$\boxed{S_k=(-1)^k\left\{k\zeta(k+1)-(k-1)\zeta(k)+\sum_{a=2}^{k-1}(a-1)(k-a)\zeta(a)\zeta(k+1-a)\right\}.} \tag{4.23}$$

The last sum is empty for $k=2$. This gives finite evaluations of an infinite family of mixed polygamma sums in ordinary zeta values and products of two such values.

Proof. Write $h=v-1$ and $T=\psi'(1+h)$. The right side of (4.19) becomes

$$-\frac{1}{2}+2h-(h+h^2)T-h^3T^2,\qquad T=\sum_{j\ge 0}(-1)^j(j+1)\zeta(j+2)h^j,\quad |h|<1.$$

The coefficient of h is $2-\zeta(2)$. For $k\ge 2$, the coefficients from the two linear terms in T are $(-1)^k\{k\zeta(k+1)-(k-1)\zeta(k)\}$; the coefficient from $-h^3T^2$ is the finite convolution in (4.23). Normal convergence, proved above, identifies these coefficients with the stated sums. □

There is also a simple resonant example. Taking $p=q=d=2$ in Theorem 4.1 gives

$$\mathcal{K}_2(v)=-2\psi(v)+(1-2v)T(v)-(v-1)^2T(v)^2.$$

Here the summand has leading term $2/x_n$, and $\mathcal{K}_2(v+N) = -2\log N - 3 + o(1)$. Comparing this expansion with the finite harmonic sum proves

$$\boxed{\begin{aligned}&\sum_{n\ge 0}\left[(2n+2v-1)\psi'(n+v)^2 - \frac{2}{n+v-1/2}\right]\\ &\quad = 3 - 2\psi(v) + 2\psi(v-\tfrac12) + (1-2v)\psi'(v) - (v-1)^2\psi'(v)^2, \qquad v > \tfrac12.\end{aligned}} \tag{4.24}$$

Its Dirichlet finite part is instead

$$\mathrm{FP}_{s=0}\sum_{n\ge 0}(2n+2v-1)\psi'(n+v)^2 x_n^{-s} = 3 - 2\psi(v) + (1-2v)\psi'(v) - (v-1)^2\psi'(v)^2. \tag{4.25}$$

The difference is exactly $2\gamma_0(v-1/2)$ as prescribed by Theorem 4.3. In particular the ordinary sum in (4.24) at $v=1$ is $3-4\log 2-\zeta(2)$, whereas its continued finite part is $3+2\gamma-\zeta(2)$.

For a convergent higher-order example requiring no subtraction, $p=q=4$, $d=5$ gives

$$\sum_{n\ge 0}\{(n+v)^5 - (n+v-1)^5\}\zeta(4,n+v)^2 = \zeta(2,v) + v(1-v)\zeta(4,v) - (v-1)^5\zeta(4,v)^2, \qquad v>0. \tag{4.26}$$

These formulas are exact finite reductions in single Hurwitz values and endpoint products. They do not require treating a nonzero formal rank or a small numerical residual as an arithmetic independence claim.

5 An alternating two-parameter digamma identity

The centered moments in the preceding sections use even polynomial weights. An alternating outer sign has a complementary effect: odd centered weights lead to odd-weight alternating double zeta values, which admit a classical parity reduction. Summing these reductions over both even tail orders gives a compact identity in digamma and trigonometric functions. The method of turning products of tails into ordered sums is classical; see Hoffman [9]. The parity formula is due to the classical Euler-sum theory and is proved in a generating-function form by Teo [10]; see also the descending-index formulation in [11, Eq. (3.7)]. The contribution here is the resulting two-parameter identity, its explicitly normalized odd-weight family, and the rational-parameter square evaluations. No literature-priority claim is made for individual special cases.

For $\sigma \in \{1,-1\}$ set

$$E_n^{\sigma}(x) = \sum_{k>n}\frac{\sigma^k}{k^2-x^2}, \qquad x \notin \mathbb{Z}\setminus\{0\}, \quad n \ge 0. \tag{5.1}$$

Each E_n^{σ} is even in x. Its poles at integers of modulus at most n are absent. These functions are ordinary polygamma expressions:

$$\begin{aligned}E_n^{+}(x) &= \frac{\psi(n+1+x)-\psi(n+1-x)}{2x},\\ E_n^{-}(x) &= \frac{(-1)^{n+1}}{2x}\{\mathcal{B}(n+1-x) - \mathcal{B}(n+1+x)\}, \qquad \mathcal{B}(v) = \frac{\psi((v+1)/2)-\psi(v/2)}{2}.\end{aligned} \tag{5.2}$$

The apparent singularities at $x=0$ are removable. Formula (5.2) follows by partial fractions and the convergent digamma difference series; the second formula follows from $\mathcal{B}(v) = \sum_{j\ge 0}(-1)^j/(j+v)$ for $\Re v > 0$, and then by continuation.

Define the even meromorphic functions

$$O_+(t) = -\frac{\psi(1+t)+\psi(1-t)+2\gamma}{2}, \tag{5.3}$$
$$O_-(t) = -\log 2 + O_+(t/2) - O_+(t),$$

and put $R_+(t) = \cot(\pi t)$ and $R_-(t) = \csc(\pi t)$. For $xy \neq 0$, abbreviate

$$U_\sigma(x,y) = \frac{O_\sigma(x+y) - O_\sigma(x-y)}{2xy}, \qquad V_\sigma(x,y) = \frac{O_\sigma(x+y) + O_\sigma(x-y)}{2}. \tag{5.4}$$

Theorem 5.1 (The two-parameter alternating identity)**.** *Let $\alpha, \beta \in \{1,-1\}$ and $c = -\alpha\beta$. Initially assume $|x| + |y| < 1$ and $xy \neq 0$. Then*

$$\begin{aligned}\mathcal{K}_{\alpha,\beta}(x,y) &:= \sum_{n=0}^{\infty} (-1)^n (2n+1) E_n^\alpha(x) E_n^\beta(y) \\ &= \frac{\pi}{2}\Big\{ \frac{R_\alpha(x)}{x} O_{-\beta}(y) + \frac{R_\beta(y)}{y} O_{-\alpha}(x) - yR_c(y)U_\alpha(x,y) - xR_c(x)U_\beta(x,y) \\ &\qquad - \frac{R_c(x)}{x} V_{-\beta}(x,y) - \frac{R_c(y)}{y} V_{-\alpha}(x,y)\Big\}. \end{aligned} \tag{5.5}$$

The series converges absolutely and locally normally if at least one of α, β is -1. For $\alpha = \beta = 1$ it converges ordinarily and locally uniformly; after subtracting each summand's value at $(0,0)$ the remaining series is absolutely locally normally convergent. Both sides extend meromorphically to $\mathbb{C}^2$. Possible genuine poles occur only when x or y is a nonzero integer. In particular the apparent singularities at $xy = 0$, and those on $x \pm y \in \mathbb{Z}$ away from the coordinate poles, are removable. The identity is even separately in x and y and is unchanged by simultaneously exchanging (α, x) with (β, y).

5.1 An ordinary finite-cutoff identity

For $s \geq 2$, write

$$T_s^\sigma(n) = \sum_{k>n} \frac{\sigma^k}{k^s}, \qquad Z(r,s;\sigma,\tau) = \sum_{k>m\geq 1} \frac{\sigma^k \tau^m}{k^r m^s}.$$

The convention for Z is the decreasing-index convention. Put $\mathsf{L}_s(\sigma) = \mathrm{Li}_s(\sigma)$ for $s \geq 2$ and $\mathsf{L}_1(-1) = -\log 2$.

Lemma 5.2 (Odd alternating weights)**.** *Let $p, q \geq 2$ be integers and let d be odd with $1 \leq d < \min(p,q)$. Then*

$$\begin{aligned}&\sum_{n\geq 0} (-1)^n \{n^d + (n+1)^d\} T_p^\alpha(n) T_q^\beta(n) \\ &\quad = -Z(p, q-d; \alpha, -\beta) - Z(q, p-d; \beta, -\alpha) - \mathsf{L}_{p+q-d}(-\alpha\beta). \end{aligned} \tag{5.6}$$

The left side is absolutely convergent except possibly when $(p,q,d,\alpha,\beta) = (2,2,1,1,1)$; in that case it converges as an ordinary alternating series. Every double series on the right is absolutely convergent.

Proof. For $m \geq 1$, cancellation of consecutive powers gives

$$\sum_{n=0}^{m-1} (-1)^n \{n^d + (n+1)^d\} = (-1)^{m-1} m^d.$$

First sum over the finite range $0 \le n \le N$. For each fixed n the two tail series are absolutely convergent, so their product may be expanded and the finite n-sum interchanged. The inner weight becomes $(-1)^{\min(k,m,N+1)-1}\min(k,m,N+1)^d$. The limiting double series is dominated by $\min(k,m)^d/(k^p m^q)$, whose sum is finite because $q-d, p-d \ge 1$ and $p,q \ge 2$. For example the region $k > m$ is

$$\sum_{m\ge 1} m^{d-q} \sum_{k>m} k^{-p} = O\left(\sum_{m\ge 1} m^{d+1-p-q}\right) < \infty.$$

Dominated convergence is therefore applicable to the double series. Separating $k > m$, $m > k$, and $k = m$ proves (5.6). Equivalently, the additional term in the finite-cutoff formula when both indices exceed $N+1$ has modulus at most $(N+1)^d T_p^+(N) T_q^+(N) = O(N^{d+2-p-q})$, which tends to zero.

For absolute convergence of the original series use $T_s^+(n) = O(n^{1-s})$ and $T_s^-(n) = O(n^{-s})$. Under the stated range the only unresolved exponent is -1 in the listed case. There $(2n+1)T_2^+(n)^2 = 2/n + O(n^{-2})$ for $n \ge 1$, so an ordinary alternating harmonic series plus an absolutely convergent series proves convergence. This also fixes the boundary value without assigning a value to a divergent series. □

5.2 The classical parity input, with its endpoint convention

The following formula is the alternating double-Euler parity theorem [10, 11]. We record it in the convention used here. For $r \ge 2$, $s \ge 1$, and $w = r + s$ odd, set

$$\mu_j = (-1)^r \left\{ \binom{j-1}{r-1} \mathsf{L}_j(\sigma) + \binom{j-1}{s-1} \mathsf{L}_j(\tau) \right\}.$$

Binomial coefficients outside $0 \le k \le n$ are zero. Then

$$\begin{aligned} Z(r,s;\sigma,\tau) = \frac{1}{2}\left\{-\mathsf{L}_w(\sigma\tau) + (1+(-1)^r)\mathsf{L}_r(\sigma)\mathsf{L}_s(\tau) + \mu_w\right\} \\ - \sum_{k=1}^{(w-1)/2} \mathsf{L}_{2k}(\sigma\tau)\mu_{w-2k}. \end{aligned} \tag{5.7}$$

The exceptional notation $\mathsf{L}_1(1)$ is interpreted as one formal symbol T while this finite expression is assembled. It cancels before the expression is evaluated. Indeed it can occur only when $s = 1$ and $\tau = 1$; then r is even, and the first product contributes $\mathsf{L}_r(\sigma)T$, while the last summand contributes its negative. There is no remaining regularization parameter. Thus one may set $\mathsf{L}_1(1) = 0$ in a coefficient implementation after this cancellation has been established. The value zero is a bookkeeping convention, not an asserted value of the divergent harmonic series.

Combining (5.6) and (5.7) already proves an explicit all-order family whenever $p+q$ is even. Only single zeta values and $\log 2$ remain, since $\mathsf{L}_s(-1) = (2^{1-s}-1)\zeta(s)$ for $s \ge 2$. No period-independence assumption is involved. The canonical weights exhaust the corresponding odd polynomial space: for odd d, $P_d(X) = (X-1/2)^d + (X+1/2)^d$ is odd with leading term $2X^d$. Consequently the P_d with $d < \min(p,q)$ form a triangular basis for all odd centered polynomial weights in that degree range. For $p+q$ even, every such weighted moment therefore has the same finite reduction to single zeta values and $\log 2$.

5.3 Proof of the two-parameter identity

Put

$$A_\sigma(t) = \sum_{a\ge 1} \mathsf{L}_{2a}(\sigma) t^{2a-2} = \frac{1}{2t^2} - \frac{\pi R_\sigma(t)}{2t}.$$

The equality follows from the cotangent partial-fraction expansion and its alternating counterpart. Moreover, for $|t| < 1$,

$$O_\sigma(t) = \sum_{j\ge 0} \mathsf{L}_{2j+1}(\sigma) t^{2j},$$

with the already justified convention $\mathsf{L}_1(1) = 0$. The familiar digamma Taylor expansion gives (5.3).

We first verify the analytic legitimacy of the coefficient calculation. Uniformly for x in a compact subset avoiding the coordinate poles,

$$E_n^+(x) = O(n^{-1}), \quad E_n^-(x) = O(n^{-2}), \quad E_n^\sigma(x) - E_n^\sigma(0) = O(n^{-3}).$$

The last estimate follows directly by subtracting the two rational summands in (5.1). Consequently, if one sign is negative, the defining series for $\mathcal{K}$ converges absolutely normally. In the positive-positive case its difference from the series at $(0,0)$ is absolutely normally convergent, with summands $O(n^{-3})$. The scalar series at $(0,0)$ converges by Lemma 5.2. Cauchy's coefficient formula therefore gives

$$[x^{2a-2}y^{2b-2}]\mathcal{K}_{\alpha,\beta} = -Z(2a, 2b-1; \alpha, -\beta) - Z(2b, 2a-1; \beta, -\alpha) - \mathsf{L}_{2a+2b-1}(c). \tag{5.8}$$

Insert (5.7). Its two terms $-\mathsf{L}_{2a+2b-1}(c)/2$ cancel the diagonal term in (5.8). Summation of the remaining binomial terms uses only the even and odd parts of the binomial theorem. For $N \ge 1$,

$$\sum_{a=1}^{N} \binom{2N}{2a-1} x^{2a-2} y^{2N-2a} = \frac{(x+y)^{2N} - (x-y)^{2N}}{2xy},$$
$$\sum_{a=0}^{N} \binom{2N}{2a} x^{2a} y^{2N-2a} = \frac{(x+y)^{2N} + (x-y)^{2N}}{2}.$$

For completeness, the complete expression before replacing A_σ by trigonometric functions is

$$\begin{aligned}\mathcal{K}_{\alpha,\beta} = {} & \left(y^2 A_c(y) - \tfrac{1}{2}\right) U_\alpha + \left(x^2 A_c(x) - \tfrac{1}{2}\right) U_\beta \\ & + \left(A_c(x) - \tfrac{1}{2x^2}\right) V_{-\beta} + \left(A_c(y) - \tfrac{1}{2y^2}\right) V_{-\alpha} \\ & + \left(\tfrac{1}{2x^2} - A_\alpha(x)\right) O_{-\beta}(y) + \left(\tfrac{1}{2y^2} - A_\beta(y)\right) O_{-\alpha}(x).\end{aligned} \tag{5.9}$$

All the power series used here converge absolutely when $|x| + |y| < 1$. Substitution of A_σ proves (5.5). If the formal constant T is retained in O_+, it cancels directly in this last expression: when $-\beta = 1$ one has $c = \alpha$, and when $-\alpha = 1$ one has $c = \beta$.

The left side is holomorphic on $(\mathbb{C} \setminus (\mathbb{Z} \setminus \{0\}))^2$ by the same normal convergence argument; near a coordinate pole only finitely many tail terms are singular. The right side is meromorphic on $\mathbb{C}^2$. The meromorphic identity theorem extends their equality from the initial open set. This proves all the removability and continuation claims in Theorem 5.1. Evenness and exchange symmetry follow directly from the left side.

There is also a direct check of the coordinate axes. Pair the first and fifth terms in the braces of (5.5) as

$$\frac{R_\alpha(x) - R_c(x)}{x} O_{-\beta}(y) + \frac{R_c(x)}{x}\{O_{-\beta}(y) - V_{-\beta}(x,y)\}.$$

All R_σ have the same principal part $1/(\pi x)$, so the first quotient is holomorphic at zero. The second difference is divisible by x^2, so the second product is also holomorphic. The symmetric pair handles $y = 0$. Finally, the difference of any even power evaluated at $x+y$ and $x-y$ is divisible by xy, which proves that U_σ is holomorphic on both axes.

5.4 A diagonal square with elementary rational-parameter values

Corollary 5.3 (A closed alternating square)**.** *For every $x \notin \mathbb{Z} \setminus \{0\}$,*

$$\boxed{\sum_{n\geq 0}(-1)^n(2n+1)\left(\sum_{k>n}\frac{(-1)^k}{k^2-x^2}\right)^2 = -\frac{\pi\{\psi(1+x)+\psi(1-x)+2\gamma\}}{4x\sin(\pi x)}.} \tag{5.10}$$

The series is absolutely locally normally convergent. Its value at $x=0$ is $\zeta(3)/2$.

Proof. Set $\alpha=\beta=-1$ and $y=x$ in (5.5). Here $c=-1$, $O_+(0)=0$, and

$$O_-(2x)-O_-(0)=O_+(x)-O_+(2x).$$

The two terms at $2x$ cancel, leaving $\pi O_+(x)/(2x\sin\pi x)$, which is (5.10). At zero, $O_+(x)=\zeta(3)x^2+O(x^4)$ and $x\sin(\pi x)=\pi x^2+O(x^4)$. This proves the value and the removable interpretation. □

The complete diagonal can be written as one symmetric matrix. With $c_x=\cos(\pi x)$ and the signs ordered as $+,-$,

$$\left(\mathcal{K}_{\alpha,\beta}(x,x)\right)_{\alpha,\beta\in\{+,-\}} = \frac{\pi}{2x\sin(\pi x)}\begin{pmatrix} 2c_xO_-(x)-O_+(x)+2\log 2 & O_-(x)+c_x\log 2 \\ O_-(x)+c_x\log 2 & O_+(x)\end{pmatrix}. \tag{5.11}$$

This follows from the same substitution $y=x$ and $O_+(2x)+O_-(2x)=O_+(x)-\log 2$ in the other sign cases. In particular,

$$\left(\mathcal{K}_{\alpha,\beta}(1/2,1/2)\right)_{\alpha,\beta} = \pi\begin{pmatrix} 1 & -1 \\ -1 & 2\log 2-1\end{pmatrix}. \tag{5.12}$$

Here $E_n^+(1/2)=1/(n+1/2)$, so the positive-positive entry also reduces directly to the ordinary alternating arctangent series for π. This example lies on the apparent oblique polar line $x+y=1$ of the unsimplified master formula, and makes its removability explicit.

For $x=r/N$, $1\leq r<N$, Gauss's digamma formula [15, Eq. (5.4.19)] gives the explicit finite expression

$$O_+(r/N)=\log(2N)-\frac{N}{2r}-2\sum_{j=1}^{\lfloor (N-1)/2\rfloor}\cos\frac{2\pi rj}{N}\log\sin\frac{\pi j}{N}. \tag{5.13}$$

Thus every such square is π times an explicit linear combination, with algebraic coefficients, of 1 and logarithms of positive algebraic numbers. For example,

$$\sum_{n\geq 0}(-1)^n(2n+1)\left(\sum_{k>n}\frac{(-1)^k}{k^2-1/4}\right)^2 = \pi(2\log 2-1), \tag{5.14}$$

$$\sum_{n\geq 0}(-1)^n(2n+1)\left(\sum_{k>n}\frac{(-1)^k}{k^2-1/9}\right)^2 = \frac{3\sqrt{3}\pi}{2}(\log 3-1), \tag{5.15}$$

$$\sum_{n\geq 0}(-1)^n(2n+1)\left(\sum_{k>n}\frac{(-1)^k}{k^2-1/16}\right)^2 = 2\sqrt{2}\pi(3\log 2-2). \tag{5.16}$$

The squares are algebraic squares throughout; no complex conjugation is inserted when x is complex.

Expanding (5.10) produces a short all-order convolution. With $\eta(s) = (1 - 2^{1-s})\zeta(s)$ for $s > 1$, for every $m \geq 2$ one has

$$\sum_{a=1}^{m-1} \sum_{n\geq 0} (-1)^n (2n+1) T_{2a}^{-}(n) T_{2m-2a}^{-}(n) = \frac{1}{2}\zeta(2m-1) + \sum_{j=1}^{m-2} \eta(2j)\zeta(2m-2j-1). \tag{5.17}$$

Indeed $\pi x/\sin \pi x = 1 + 2\sum_{j\geq 1} \eta(2j)x^{2j}$; coefficient extraction is legitimate by absolute normal convergence.

Corollary 5.4 (An anchored Gamma primitive)**.** *For $|u| < 1$,*

$$\int_0^u x \sin(\pi x) \mathcal{K}_{-,-}(x,x)\, dx = -\frac{\pi}{4} \left\{ \log \Gamma(1+u) - \log \Gamma(1-u) + 2\gamma u \right\}. \tag{5.18}$$

The logarithms are their analytic branches on the unit disk, normalized by $\log \Gamma(1) = 0$. The path of integration may be any path in that disk, and the defining series for $\mathcal{K}_{-,-}$ may be integrated term by term. In particular the value at $u = 1/2$ is $\pi(\log 2 - \gamma)/4$.

Proof. Multiply (5.10) by $x \sin(\pi x)$. Its right side is the derivative of the expression in braces in (5.18), multiplied by $-\pi/4$. The anchored logarithms exist because $\Gamma(1 \pm u)$ are holomorphic and nonzero on the simply connected unit disk. Both sides vanish at zero. Absolute local normal convergence, already proved, justifies exchanging the path integral and sum. Finally $\Gamma(3/2)/\Gamma(1/2) = 1/2$ gives the stated value. □

Some individual coefficients of the full two-variable theorem are also useful. Let $C_{p,q}^{\alpha,\beta} = \sum_{n\geq 0} (-1)^n (2n+1) T_p^{\alpha}(n) T_q^{\beta}(n)$. Then

$$C_{2,2}^{+,+} = \frac{\pi^2 \log 2}{2} - \frac{5}{4}\zeta(3), \qquad C_{2,2}^{+,-} = -\frac{\pi^2 \log 2}{4} - \frac{3}{8}\zeta(3), \tag{5.19}$$

$$C_{2,2}^{-,-} = \frac{1}{2}\zeta(3), \qquad C_{2,4}^{+,+} = \frac{\pi^4 \log 2}{48} + \frac{\pi^2 \zeta(3)}{12} - \frac{23}{32}\zeta(5), \tag{5.20}$$

$$C_{4,4}^{+,+} = -\frac{335}{64}\zeta(7) + \frac{13\pi^2}{48}\zeta(5) + \frac{\pi^4}{32}\zeta(3). \tag{5.21}$$

Only $C_{2,2}^{+,+}$ is conditionally convergent. As an example with a higher odd centered weight, (5.6) gives the absolutely convergent identity

$$\sum_{n\geq 0} (-1)^n \{n^3 + (n+1)^3\} T_4^{+}(n)^2 = \frac{\pi^4 \log 2}{24} + \frac{\pi^2 \zeta(3)}{8} - \frac{49}{16}\zeta(5). \tag{5.22}$$

5.5 All repeated rational resolvents

The same identity supplies explicit derivatives of every order. Define

$$E_{n,r}^{\sigma}(x) = \sum_{k>n} \frac{\sigma^k}{(k^2 - x^2)^{r+1}}, \qquad r \geq 0,$$

and let $\partial_{x^2} = (2x)^{-1}\partial_x$ on even analytic functions, with its removable meaning at $x = 0$.

Corollary 5.5 (Repeated-resolvent products)**.** *For integers $r, s \geq 0$ and parameters outside the coordinate poles,*

$$\sum_{n\geq 0} (-1)^n (2n+1) E_{n,r}^{\alpha}(x) E_{n,s}^{\beta}(y) = \frac{1}{r!s!} \partial_{x^2}^r \partial_{y^2}^s \mathcal{K}_{\alpha,\beta}(x,y). \tag{5.23}$$

The right side is an explicit finite expression obtained by differentiating (5.5). Whenever $r + s > 0$, the series is absolutely locally normally convergent for all sign pairs.

Proof. Termwise differentiation of the individual rational tails gives $\partial_{x^2}^r E_n^\alpha(x) = r! E_{n,r}^\alpha(x)$. For compact parameter sets avoiding poles, the positive-sign estimate is $E_{n,r}^+(x) = O(n^{-2r-1})$; an alternating sign improves it to $O(n^{-2r-2})$. Thus a derivative of positive total order makes the outer series absolutely summable, even when both signs are positive. The locally uniform identity in Theorem 5.1 and Cauchy's differentiation formula justify the operation. Evenness gives the stated removable interpretation at the axes. □

5.6 A colored polylogarithmic function and its anchored primitive

The same finite-difference mechanism has an interior-variable form that also fixes an antiderivative. For $p, q \geq 2$, $|a|, |b| \leq 1$, and $|z| < 1$, put $T_p^a(n) = \mathrm{Li}_p(a) - \sum_{k=1}^n a^k/k^p$. Use the colored convention

$$\mathrm{Li}_{s_1,\dots,s_d}(z_1,\dots,z_d) = \sum_{n_1>\cdots>n_d\geq 1} \frac{z_1^{n_1}\cdots z_d^{n_d}}{n_1^{s_1}\cdots n_d^{s_d}}.$$

Proposition 5.6 (Colored generating function and primitive)**.** *Writing $F(z) = \sum_{n\geq 0} T_p^a(n) T_q^b(n) z^n$, one has*

$$\begin{aligned}(1-z)F(z) = {} & \mathrm{Li}_p(a)\,\mathrm{Li}_q(b) - \mathrm{Li}_q(b)\,\mathrm{Li}_p(az) - \mathrm{Li}_p(a)\,\mathrm{Li}_q(bz) \\ & + \mathrm{Li}_{p,q}(az,b) + \mathrm{Li}_{q,p}(bz,a) + \mathrm{Li}_{p+q}(abz). \end{aligned} \tag{5.24}$$

The primitive normalized to vanish at zero is

$$\begin{aligned}\int_0^z F(t)\,dt = {} & \mathrm{Li}_p(a)\,\mathrm{Li}_q(b)\,\mathrm{Li}_1(z) - \mathrm{Li}_q(b)\,\mathrm{Li}_{1,p}(z,a) - \mathrm{Li}_p(a)\,\mathrm{Li}_{1,q}(z,b) \\ & + \mathrm{Li}_{1,p,q}(z,a,b) + \mathrm{Li}_{1,q,p}(z,b,a) + \mathrm{Li}_{1,p+q}(z,ab). \end{aligned} \tag{5.25}$$

The integral is along any path in the unit disk. Every function in the formulas is defined by an absolutely locally normally convergent series.

Proof. The identity

$$T_p^a(n-1)T_q^b(n-1) - T_p^a(n)T_q^b(n) = \frac{a^n}{n^p}T_q^b(n) + \frac{b^n}{n^q}T_p^a(n) + \frac{(ab)^n}{n^{p+q}}$$

proves (5.24) after multiplication by z^n and summation. All rearrangements are absolute in the disk. The general derivative

$$\frac{d}{dz}\,\mathrm{Li}_{1,s_1,\dots,s_d}(z,c_1,\dots,c_d) = \frac{\mathrm{Li}_{s_1,\dots,s_d}(zc_1,c_2,\dots,c_d)}{1-z}$$

follows by summing the largest index first. It proves the displayed primitive term by term. Each term vanishes at $z = 0$, which fixes its additive constant and branch. □

The generating function and primitive are finite-difference consequences, not assertions of general depth reduction. The additional reduction in Theorem 5.1 uses the specific alternating weight and odd-weight parity input. It does not imply the outstanding Gaussian S_6 or revised S_8 identities.

6 Source reconciliation and a further research programme

6.1 What the audit establishes

The source claims used in the proofs were checked against the pinned revision, including the finite-cutoff conventions and the exact matrices. The audit establishes one stale research-status statement and a

precise centering reconciliation. It did not establish an additional false theorem in the inspected parts of the six incoming reports. This conclusion is limited to those parts: acquiring the full source inventory does not constitute checking every proof in the collective manuscript.

Source	Material checked for this continuation	Result
Canonical fifth-moment report [2]	Tail blocks, inverse transform, diagonal rank, and the first-q conjecture	Normalization retained; gap four advances its conjecture
Differential polygamma and Lerch jets [3]	Adjacent-order proof and fixed-gap recurrence	Recurrence and determinant signs retained
Bernoulli–Tornheim–digamma [4]	Fixed smaller orders $2, 4, 6$; shifted-Hankel reduction; tail foundation; Tornheim subtraction; differential and research sections	No additional false formula found; R8 status needs reconciliation
Polygamma tail identities [5]	Moment rank and reconstruction; spectral polynomial formulas; sixth mixed moment	Supplies the already available spectral coordinates used below
Cubic tail descent [6]	Cubic Gamma product and the master identity with its subtraction	No error established in the inspected product and limit arguments
Determinantal harmonic resonance [7]	First resonance and the cutoff-to-spectral transfer law	The amplitude-derivative correction remains necessary
Multishift reflection [8]	Entire-order transform and full horizontal-cut reflection	No error established in the inspected holomorphy and jump conventions

The accompanying source-audit record identifies the exact section files and equation labels. The preserved Canonical report is already under `Analysis/Polylogarithms/docs/reports/stieltjes-harmonic-resonance/continuations/canonical-digamma-fifth/`; it is not one of the six current incoming archives. The earlier extracted tail-rank source used during this work was checked against that preserved file and has the identical Git blob.

6.2 A sixth mixed moment whose spectral evaluation already exists

Research item R8 of the Bernoulli–Tornheim–digamma report asks for a harmonic-spectral evaluation of

$$\widetilde{M}_6 = \mathrm{FP}\int_0^1 (\psi(x)+\gamma)^2\psi'(x)^2\,dx.$$

Here and throughout this subsection, $\mathrm{FP}\int_0^1 f(x)\,dx$ means the coefficient of $\varepsilon^0(\log\varepsilon)^0$ in $\int_\varepsilon^1 f(x)\,dx$ as $\varepsilon \downarrow 0$, in the original coordinate x. The operation is linear on the Laurent-log expansions used here. The existence question is already answered by the Polygamma Tail Identities report [5]. Its equation `sixth:I` evaluates the uncentered mixed integral; equations `spectral:exampleDP` and `spectral:digamma_trigammasquare` evaluate the lower mixed term. The differing centering can be reconciled explicitly.

Define, in decreasing-index convention,

$$\begin{aligned} D_P(s) = {} & -4\zeta(3,s+1) - 4\zeta(s+4) + 2\zeta(s+2,2) + 2\zeta(2)\zeta(s+2) \\ & + 4\zeta(s+3,1) - 4\gamma\zeta(s+3), \qquad \Re s > -1. \end{aligned} \tag{6.1}$$

All the displayed series are holomorphic in this half-plane. To specify the other inherited coordinate without leaving an unnamed integral, put, for $n \geq 0$,

$$b_n = H_n - \gamma, \qquad c_n = H_n^{(2)} + \zeta(2), \qquad d_n = H_n^{(3)} - \zeta(3),$$
$$e_n = H_n^{(4)} + \zeta(4), \qquad f_n = H_n^{(5)} - \zeta(5),$$

and

$$r_n = 4b_n^2 d_n + 2b_n c_n^2 - 6c_n d_n - 10 b_n e_n + 6 f_n, \tag{6.2}$$
$$\Omega_5 = -\sum_{n\geq 1} (r_n - r_{n-1}) \log n = \sum_{n\geq 1} r_n \log(1 + 1/n). \tag{6.3}$$

These are ordinary absolutely convergent series. For example, the harmonic-tail expansions and $8\zeta(2)^2 = 20\zeta(4)$ give $r_n = O(\log(n+1)/n + \log^2(n+1)/n^2)$ and $r_n - r_{n-1} = O(\log(n+1)/n^2)$; summation by parts has a vanishing boundary term. These definitions agree exactly with the source's `sixth:D` and `sixth:omega-series`.

Proposition 6.1 (Reconciliation of the centered sixth coordinate)**.** *The inherited spectral evaluations imply*

$$\boxed{\widetilde{M}_6 = \Omega_5 + 2\gamma D_P'(0) + 4\gamma^2 \zeta'(3) + \frac{31}{3}\zeta(2)\zeta(3) - \frac{19}{3}\zeta(5).} \tag{6.4}$$

Proof. The three source evaluations in the same finite-part convention are

$$\mathrm{FP}\int_0^1 \psi^2 \psi'^2 \, dx = \Omega_5 + 2\gamma^2\zeta(3) - \frac{35}{3}\gamma\zeta(4) + \frac{31}{3}\zeta(2)\zeta(3) - \frac{19}{3}\zeta(5),$$
$$\mathrm{FP}\int_0^1 \psi \psi'^2 \, dx = \frac{35}{6}\zeta(4) - 2\gamma\zeta(3) + D_P'(0),$$
$$\mathrm{FP}\int_0^1 \psi'^2 \, dx = 2\zeta(3) + 4\zeta'(3).$$

Expand $(\psi + \gamma)^2 \psi'^2$ and use linearity. The coefficients of $\gamma\zeta(4)$ and $\gamma^2\zeta(3)$ cancel exactly, leaving (6.4). In the Canonical fifth-moment notation one also has $D_P(s) = 2J(s+1)$, by the stuffle product; hence $D_P'(0) = 2J'(1)$. □

The proposed correction to R8 is therefore: determine whether the specified combination in (6.4) reduces to a smaller, explicitly stated collection of constants, such as selected ordinary zeta derivatives and prescribed height-one spectral data. Existence of a harmonic-spectral representation should be marked resolved. This reconciliation does not prove a reduction to that smaller collection, and it is not an arithmetic-independence statement about Ω_5.

6.3 Conventions that must survive integration

Three distinctions materially affect the values of the formulas. First, the original centered lattice has $\zeta(-2j, 1/2) = 0$, whereas a shifted lattice generally does not. Theorems 4.2 and 4.3 give the two required corrections explicitly. Second, equality of zero-order continued and ordinary values does not imply equality after spectral differentiation: the nonzero $\zeta^{(j)}(-k, \alpha)$ terms must be retained. Third, $\mathcal{K}_{+,+}$ and its $(2,2)$ coefficient have an ordinary conditional interpretation; their original series must not be described as absolutely convergent. The other sign pairings are absolutely convergent.

These are normalization requirements for using the new results, not claims that the inspected predecessors made all three mistakes. The separately unresolved Gaussian S_6 and revised S_8 candidates remain conjectural in this continuation. A proved reduction for a different tail family supplies no missing identity for those values.

6.4 Further research questions with concrete targets

The following questions concern exact evaluations, structure, or proof certificates. The numerical evidence accompanying the sources can help choose cases, but each proposed endpoint is an identity or a rigorous nonvanishing theorem.

Q1. Complete the optimal-prefix conjecture beyond gap four. For even $8 \le p < q$ with $q - p \ge 6$, prove or refute $\det M_{p,q} \neq 0$. The first unproved fixed-gap family is $(p, q) = (2b - 6, 2b)$ with $b \ge 7$. Its boundary determinant has size three. The explicit sign change already seen between $b = 6$ and $b = 7$ rules out a proof based on one fixed sign for this family. A useful target is an exact expression whose factors prove nonvanishing while allowing the determinant's sign to vary. Comparing the size-c fixed-gap determinant with the size-$p/2$ complementary determinant in the shifted-Hankel approach may reduce different parameter ranges.

Q2. Classify higher minors of the fundamental recurrence. Theorem 3.2 is uniform in the gap and the two chosen columns. Determine which larger minors vanish or change sign, and whether a recurrence for their exterior products separates those two possibilities. The one-negative-index induction in the present proof does not directly extend to several sign changes. A successful replacement should explain the recorded third-order sign reversal, not impose a positivity property contradicted by it.

Q3. Determine the exact formal rank after a Hurwitz shift. Section 4 gives at most $q + 2$ coordinates over $\mathbb{Q}(v)$. Is that list generically minimal, and how many initial even moments reconstruct it? Describe every exceptional shift in a stated real or complex domain, including the known degeneration at $v = 1$. This is a question about rational coefficient matrices; it does not require independence of the specialized Hurwitz values. The Bernoulli-polynomial endpoint and the highest single-Hurwitz column should be kept throughout any determinant computation.

Q4. Allow different shifts in the two factors. Replace the product by $\zeta(p, n + v)\zeta(q, n + w)$. Integer values of $v - w$ can be related by finite tail corrections. For general rational differences, determine an explicit finite normal form and the necessary depth-two shifted coordinates. The first test is whether a discrete primitive analogous to Theorem 4.1 closes in single Hurwitz values and endpoint products, and exactly which shift differences permit that closure.

Q5. Combine tail-order derivatives with the weight jets. Theorem 4.3 differentiates the factor x_n^{-s}. Derivatives in p or q instead create logarithmic harmonic tails. Construct the mixed Taylor-Laurent expansion in all three variables, including motion of the polar hyperplanes and coincidences of resonances. The target is a finite expression in specified derivatives of single and multiple Hurwitz functions and generalized Stieltjes constants, with the scale and amplitude corrections visible at every order. This would extend the source's Lerch and Tornheim jet machinery to the shifted product setting.

Q6. Replace signs by higher roots of unity. For a fixed conductor N, study the analogues of $\mathcal{K}_{\alpha,\beta}$ with Nth roots of unity as colors and a corresponding outer character. State explicitly which conjugation and weight-parity sectors reduce in depth. The level-two parity formula used here cannot simply be reused unchanged. A concrete objective is a conductor-four generator expressed through digamma at quarter shifts and a specified collection of cyclotomic polylogarithms. Such a result would give an exact bridge toward the Gaussian setting without assuming the outstanding Gaussian identities.

Q7. Find compact normal forms for differentiated resolvents. Parameter differentiation of Theorem 5.1 already gives sums of products of arbitrary repeated rational resolvents. Can their right sides be organized by finite Bell-polynomial formulas that display the removable singularities before specialization? At rational parameters, seek explicit formulas in algebraic coefficients, logarithms, and Dirichlet L-values arising from polygamma at rational arguments. A separate useful target is a direct contour or hypergeometric proof of the diagonal square and its Gamma primitive, independent of the general parity theorem.

Q8. Sum the higher odd weights into one generating identity. The finite reductions in (5.6) handle every admissible odd polynomial weight. Introduce a third variable for that degree and determine a compact generating function, or an explicit differential operator on $O_{\pm}$. The admissible degree condition must be built into the formulation: extending it crosses new convergence thresholds and requires additional subtractions. An exact endpoint prescription is part of the desired identity.

Q9. Extend the discrete primitive to higher products. For three or more Hurwitz tails, finite differences introduce products of fewer tails. Determine which polynomial weights allow this hierarchy to terminate in single Hurwitz values, and which force genuine multiple-Hurwitz coordinates. Hoffman's higher-product decomposition [9] supplies the ordered-sum framework. A useful initial problem is a complete classification for three equal even orders and weights of degree below the first resonance.

Q10. Reduce the reconciled sixth spectral coordinate. Choose a target collection first, then test whether $\Omega_5 + 2\gamma D_P'(0) + 4\gamma^2\zeta'(3)$ belongs to its rational span. Candidate collections can include specified ordinary zeta derivatives and the height-one slopes already defined by the Gamma bridge. The source gives convergent series and a finite multiple-zeta derivative representation for the current coordinate; the missing step is a proved functional reduction among those representations. Numeric relation searches alone cannot decide it.

Q11. Connect the colored primitive to endpoint periods. Proposition 5.6 fixes both an analytic function and its additive constant in the disk. Continue it to selected boundary points using a stated path, compute its endpoint limits, and compare the resulting depth-three polylogarithms under shuffle and stuffle relations. Determine which combinations admit a Gamma or polygamma description like the diagonal primitive. At a branch point, the desired result should include the full finite-part or limiting rule.

Q12. Produce exact certificates for the remaining numerical candidates. For the Gaussian S_6 and revised S_8 conjectures, isolate a finite algebraic, contour, or hypergeometric identity whose specialization would prove the claimed value. Separate the convergence argument from the period reduction, and state the intermediate functions with fixed branches. The finite telescoper in (3.22) illustrates the desired form of a checkable certificate for a sum; the Gaussian questions need their own certificates.

The immediate priorities are Q1, Q3, and Q6. They extend the three proved mechanisms of this article while preserving an exact target and a clear test for what would constitute a solution.

7 Reproducibility, proof status, and integration

7.1 Proofs and computational checks have different roles

The infinite-family conclusions are proved in the article. The sign comparison in Theorem 3.2, the finite Bernoulli primitive in Theorem 4.1, and the normally convergent coefficient argument in Theorem 5.1 are the respective all-order arguments. The programs reproduce exact finite instances, check normalizations by different representations, and provide numerical diagnostics for analytic specializations. No integer-relation search is used in any of the supplied verifiers.

Verification	Scope	What is compared
Gap-four determinants	$3 \le b \le 18$	Original Bernoulli matrix, recurrence determinant, and product formula
Generated minors	1,946 minors	Exact signs for gaps $2c$, $c = 2, \dots, 6$
Finite telescopers	$(4, 8), (8, 12)$	Rational-function differences and the complete constant-term limit
Shifted discrete primitives	168 triples	Generic-variable identities, $2 \le p \le q \le 8$, $q \le d \le q + 5$
Shifted transforms	$21 + 21$ checks	Inverse Bernoulli polynomials and independent finite rational endpoints
Shift derivatives	23 coefficients	The closed convolution for derivative orders $2, \dots, 24$
Alternating master	144 coefficients	Every even $p, q \ge 2$, $p + q \le 18$, for all four sign pairs
Formal endpoint cancellation	256 checks	Cancellation of the single formal $\zeta(1)$ symbol
Analytic diagnostics	$36 + 16$ checks	Independent shifted asymptotics and signed digamma-tail sums

The gap-four program also verifies the row and column scalings against direct cotangent coefficient extraction, checks the highest polynomial coefficient, and reproduces the higher-gap sign obstruction. The alternating program expands the meromorphic right side as a Laurent polynomial in two variables, verifies cancellation of axial negative powers and odd powers, and checks the diagonal convolution. The shifted program treats the relevant Hurwitz values as formal symbols obeying their exact recurrence, so its discrete-primitive checks are symbolic identities in the shift, not tests at one shift.

The shifted diagnostics use 85 decimal digits and the alternating diagnostics use 55. Each comparison has a 10^{-45} acceptance threshold. The observed residuals are below 7.1×10^{-65} and 6.6×10^{-55}, respectively, in the recorded run. These numbers are numerical consistency measurements, not certified error enclosures. The proofs, including their convergence arguments, do not depend on treating those measurements as exact zeros.

7.2 Contents of the accompanying archive

The archive includes a standalone TeX source and its compiled PDF, the modular source used for editing, all three independent verification programs, their complete JSON results, and a reproducibility record. The source manifest fixes the repository revision, paths, byte sizes, Git blob identifiers, and local SHA-256 checksums for the acquired inputs. The proof-status record distinguishes new theorems, inherited ingredients, finite computational checks, and unresolved questions. No large predecessor PDF or redundant copy of the repository is required to rebuild this article.

For verification from the archive directory, the three commands are

```
python3 verification/verify_gap_four.py
  -output results/gap_four_exact_results.json
python3 verification/verify_shifted_tails.py
  -output results/shifted_tail_results.json
python3 verification/verify_alternating_master.py -numeric
  -output results/alternating_verification.json
```

Each indented continuation belongs to the preceding shell command. The plain-text README contains directly copyable single-line commands. Python, SymPy, and mpmath are sufficient for these checks. The exact dependency versions used are recorded in the package. Rebuilding the document requires a LaTeX installation with the packages declared in its preamble; the provided build program assembles the standalone source and runs `latexmk`.

7.3 Suggested integration points

The gap-four theorem belongs beside the optimal-prefix conjecture and the existing adjacent-order and fixed-smaller-order proofs. Preserve the matrix convention in (3.3), especially the omission of a binomial coefficient with lower index zero. Record the newly settled family as $q - p = 4$ and leave Q1's remaining range open.

The Hurwitz-shift transform can follow the centered-tail foundation. Keep (4.8) and (4.18) close to the transform itself, because they are part of its value normalization. The all-jet theorem belongs with the spectral-subtraction and Lerch jet material; its derivatives act on the weight exponent.

The signed master, Gamma primitive, and colored polylogarithmic primitive form a self-contained continuation of the tail generating functions. Retain the explicit convergence distinction for the positive-positive entry. Finally, update the sixth mixed-moment research item using Proposition 6.1; retain the smaller-collection reduction problem as Q10.

These integration suggestions are supplied as reviewable material. The repository itself was not modified during this research.

References

[1] Vladimir Reshetnikov, maintainer, *ProveIt*, repository snapshot `446dca9bd65c20e409d1d2561d4ffad408103e2a`, 11 October 2026. Polylogarithm manuscript sources; six incoming reports. The accompanying manifest identifies every acquired source blob.

[2] ProveIt research reports, *Canonical Digamma and Fifth Moment*, preserved `canonical-digamma-fifth` continuation, 11 October 2026. In particular, `sections/07_tail_rank.tex` and its optimal-prefix conjecture. Preserved report at the pinned revision.

[3] ProveIt research reports, *Differential Polygamma and Lerch Identities*, 11 October 2026. In particular, `sections/08_adjacent_tails.tex`. Incoming source archive.

[4] ProveIt research reports, *Bernoulli Determinants, Cubic Tornheim Jets, and Canonical Digamma Integrals*, 11 October 2026. The tail foundation, general shifted-Hankel reduction, fixed smaller orders four and six, and research item R8. Incoming source archive.

[5] ProveIt research reports, *Exact Polylogarithm Parity, Harmonic-Tail Reconstruction, and Higher Polygamma Moments*, 11 October 2026. In particular, `sections/07_spectral_polynomials.tex` and `sections/08_sixth_moment.tex`. Incoming source archive.

[6] ProveIt research reports, *Cubic Invariant Tail Descent*, 10 October 2026. Cubic Gamma product and master identity with explicit subtraction. Incoming source archive.

[7] ProveIt research reports, *Determinantal Harmonic Resonance*, 10 October 2026. First resonance and finite-part transfer; archive root `ProveIt_Determinantal_Harmonic_Resonance/`. Incoming source archive.

[8] ProveIt research reports, *Contact-Complete Multishift Lerch Reflection*, 10 October 2026. Entire-order transform and horizontal cuts. Incoming source archive.

[9] M. E. Hoffman, *Sums of Products of Riemann Zeta Tails*, Mediterranean Journal of Mathematics **13** (2016), 2771–2781. https://arxiv.org/abs/1602.03399. https://doi.org/10.1007/s00009-015-0653-9.

[10] L.-P. Teo, *Alternating Double Euler Sums, Hypergeometric Identities and a Theorem of Zagier*, Journal of Mathematical Analysis and Applications **462** (2018), 777–800. https://arxiv.org/abs/1705.01269. https://doi.org/10.1016/j.jmaa.2018.02.037.

[11] C. Xu and W. Wang, *Explicit formulas of Euler sums via multiple zeta values*, arXiv:1805.08056v3, 2019, equation (3.7). https://arxiv.org/abs/1805.08056.

[12] K. Dilcher and L. Jiu, *Hankel Determinants of Sequences Related to Bernoulli and Euler Polynomials*, arXiv:2007.09821, 2020. https://arxiv.org/abs/2007.09821.

[13] NIST Digital Library of Mathematical Functions, *Hurwitz Zeta Function*, Section 25.11. Recurrence, polygamma relation, negative integer values, and Euler–Maclaurin representations. https://dlmf.nist.gov/25.11.

[14] NIST Digital Library of Mathematical Functions, *Bernoulli and Euler Polynomials: Basic Properties*, Section 24.4. https://dlmf.nist.gov/24.4.

[15] NIST Digital Library of Mathematical Functions, Sections 4.22, 5.4, 5.7, and 5.15. Cotangent partial fractions, Gauss's digamma formula, Taylor series, and polygamma identities. https://dlmf.nist.gov/4.22; https://dlmf.nist.gov/5.4; https://dlmf.nist.gov/5.7; https://dlmf.nist.gov/5.15.